

The Dataset Operations Platform.

A complete technical blueprint for designing, building and operating the software platform behind Caudals — the B2B marketplace and managed services layer for AI-ready, compliant, ML-grade datasets.

PREPARED FOR

Caudals Founding Team
Engineering · Operations ·
Strategy

SCOPE

Dataset operations ·
Marketplace
Internal admin · Delivery

CLASSIFICATION

Internal · Source-of-truth
Supersedes prior drafts

A blueprint for the operational layer of B2B AI data.

This document is the canonical technical specification for the Caudals platform. It describes the system architecture, technology stack, data operations pipeline, product surfaces, internal tooling, and operating model required to source, build, and deliver multi-modal AI datasets at professional B2B quality. Every implementation decision in this repository should trace back to a clause in this document.

Audience

Caudals founders, platform engineers, data engineers, ML engineers, data scientists, operations and labeling reviewers, security and compliance leads, and any external collaborators contracted to extend the platform.

Reading order

1. Executive summary & the operating model.
2. System architecture and technology stack.
3. The dataset operations pipeline (Part II) — the core of the business.
4. Product surfaces (Part III) and internal operations (Part IV).
5. Implementation roadmap and risk register (Part V).

Out of scope

Marketing copy, pricing rate cards, individual customer integrations, and specific commercial contracts. Those live in adjacent documents and change at higher cadence.

DOCUMENT CONTROL

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CLASSIFICATION	Internal · do not distribute externally

CONVENTIONS

MUST / SHOULD / MAY follow RFC 2119. Code, paths and identifiers appear in `monospace`. Names of external products are cited only as references; final selection happens in §05 (Technology Stack). Diagrams are illustrative; the source of truth for schema and APIs is the

implementation, kept in sync via contract tests described in §29.

5

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DATA MODALITIES

Five parts. Thirty-four sections. One platform.

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What we are building, and why it has to exist now.

AI teams are running out of usable data, while data-rich companies sit on proprietary assets they cannot safely sell. Caudals exists to close that gap. This blueprint describes the platform required to do so professionally — at the level of rigour expected of regulated B2B infrastructure, not a consultancy with spreadsheets.

The thesis

The market needs a trusted operational layer between data suppliers and AI buyers. Marketplaces alone are listing directories; labeling vendors alone do not solve sourcing or licensing; consultancies do not scale. Caudals combines marketplace gravity, supplier-side monetization, managed dataset engineering, and rights/provenance/privacy review into a single platform with a published catalogue and bespoke build capacity.

The platform

The Caudals platform is structured around three concentric rings:

- 01 Ring 1 — Public surface.** Landing, blog, contact, and demand-capture APIs. Already live; the front door for buyers and suppliers.
- 02 Ring 2 — Internal operations.** The admin console where operators run the business: leads, supplier assets, builds, QA, contracts, deliveries, and audit history.
- 03 Ring 3 — Dataset Operations Platform.** The engineering core: a medallion-architected pipeline that turns raw supplier data into ML-ready, licensed, audited dataset artefacts across eight modalities.

THE ONE-LINE PRODUCT

A managed B2B platform that turns proprietary company data into licensed, ML-ready datasets, with auditable provenance, privacy handling and quality scoring — sold privately or through a curated catalogue.

HARD CONSTRAINTS

- **Provenance is non-negotiable.** Every record must trace to a source, a contract, and a permitted-use clause.
- **Privacy by default.** PII is detected at ingest, not at delivery.
- **Licensing is data, not paper.** Permitted uses, derivative rights, exclusivity, geography and term live on the asset, not in a PDF.
- **Builds are reproducible.** Any released dataset version can be rebuilt byte-for-byte from its lineage manifest.

Where the moat is

The competitive moat is not the labeling UI or the marketplace listing page. It is the *operations spine*: a unified data model for assets, rights, builds, runs, labels, lineage, QA, contracts and deliveries; a programmable pipeline that composes profiling, cleaning, privacy, enrichment, labeling, QA and packaging as reusable assets; and an audit trail strong enough to satisfy enterprise procurement, EU AI Act Article 10 documentation, and supplier revenue-share contracts.

Build sequencing

The roadmap (§30) commits to a 90-day, 6-month and 12-month sequence. The 90-day target is operator-grade: an admin console capable of running five concurrent dataset builds end-to-end with full lineage and licensed delivery. Buyer and supplier self-service ship after operations are reliable.

- **Modalities are first-class.**
Tabular, text, image, video, audio, geospatial, time-series and document datasets share one platform.

NORTH-STAR METRICS

Time-to-pilot	≤ 14 days
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Build reproducibility	100%
------------------------------	------

PII leak incidents	0 / quarter
---------------------------	----------------

License audit pass	≥ 99%
---------------------------	-------

Avg QA score released	≥ 0.85
------------------------------	--------

"The goal is not to be a listing directory, a labeling provider, or a consulting shop. The goal is to be the trusted transaction and operations layer for AI-ready B2B datasets." — Caudals overview, internal.

PART I

Strategic & architectural foundation.

Five sections establishing why Caudals exists, how its operating model works, and the architectural principles every implementation choice in this document descends from. Read first; everything else assumes it.

01

Mission & thesis

02

Operating model

03

Principles

04 · 05

Architecture & stack

A platform between the data-rich and the model-hungry.

The strategic premise: AI capability has decoupled from training data quality. Compute is plentiful, models are commoditized, and yet most domain-specific AI fails because the underlying datasets are stale, legally ambiguous, biased, or simply not granular enough. Meanwhile, operationally rich companies sit on data that would be transformative for a buyer two industries away — but they have no compliant path to sell it.

The market gap, precisely

What buyers cannot get

- Domain-specific training and evaluation data with documented provenance.
- Datasets legally cleared for AI training, fine-tuning and commercial inference.
- ML-ready formats with verifiable PII handling and bias documentation.
- Custom builds with feasibility memos, samples, schemas and QA scorecards *before* commitment.
- Recurring data feeds with versioning, lineage, and a refresh contract.

What suppliers cannot do alone

- Stand up a sales pipeline aimed at AI buyers they cannot identify.
- Run a privacy / consent / DSAR pipeline for derivative datasets.
- Negotiate AI-training licenses without external counsel.
- Package data into Parquet, JSONL, COCO, YOLO, TFRecords, WebDataset.
- Operate a marketplace, payments rail, and support function around data products.

The Caudals thesis

Caudals occupies the empty quadrant in this map: a managed operations layer with marketplace gravity. Three things happen on the platform that do not happen anywhere else in one place:

- 01 Sourcing & rights.** We identify supplier-side proprietary data, validate AI-training and commercial-inference rights, and convert that legal posture into machine-readable license records.
- 02 Engineering.** We run the dataset operations pipeline (Part II) as a programmable system — not as a bespoke per-project script. Profiling, cleaning, privacy, enrichment, labeling, QA, packaging and delivery compose like Lego blocks per build.

03 Distribution. We package each accepted dataset as a catalogue listing or a private offer with versioning, refresh cadence, revenue share, and acceptance evidence.

What Caudals is not

- **Not a labeling vendor.** Labeling is one stage of nine.
- **Not only a marketplace directory.** Listings without operations are commodities.
- **Not a consultancy.** Bespoke project work runs through productized pipelines, not custom scripts.
- **Not a platform sold to AI labs only.** The buyer base is enterprise ML teams across regulated and unregulated verticals.

Three sides, one operations spine.

The platform serves three populations whose workflows must be modelled as distinct surfaces with distinct permissions, but which converge on a single internal system of record. Everything — lead, asset, build, contract, delivery — is one row in one connected schema described in §28.

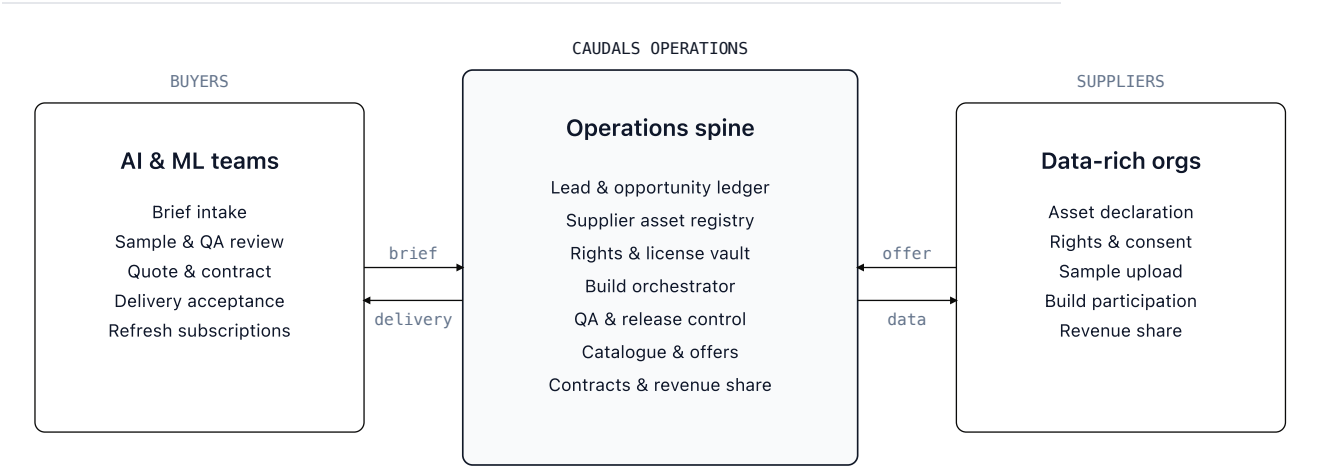


FIG 02.1 — THREE-SIDED PLATFORM WITH A SINGLE OPERATIONS SPINE.

FIG. 02.1

Buyer-side operations

Buyers enter through (a) the public contact funnel, (b) outbound sales, (c) catalogue self-service once exposed, or (d) referral. Each entry becomes a **buyer_opportunity** record that progresses through qualification, scoping, sampling, quoting, contracting, delivery and renewal. Buyer-side product surfaces are described in §20.

Supplier-side operations

Suppliers enter through (a) outbound sourcing, (b) the public contact funnel for monetization interest, or (c) referral from existing partners. Each becomes a **supplier_organization** with a registry of **supplier_asset** records, each with its own provenance, consent posture, rights declaration and dataset suitability. Supplier-side product surfaces are described in §21.

Internal operations

The Caudals operations team runs the business through a single console (§19) that exposes every record needed to qualify, build, audit, ship and bill datasets. The operating model is intentionally

operator-led for the first 12 months: self-service buyer/supplier flows ship only after the operations spine is reliable and the catalogue is populated.

Twelve principles. Every implementation choice descends from these.

These are not aspirations. They are constraints. A pull request that violates one of these without an explicit, documented exception in this blueprint should be rejected.

P-01 · Single source of truth

Every business object — lead, asset, build, run, label batch, QA report, contract, delivery — lives in exactly one canonical table. No shadow spreadsheets. Reports are derived; assertions are authoritative.

P-02 · Datasets are assets, not pipelines

We treat each materialized dataset version as a first-class asset with a manifest, a hash, a license, a lineage graph, and an immutable content URI. Pipelines produce assets; assets are the unit of release.

P-03 · Medallion layering

All data flows through **bronze** (raw + provenance stamp), **silver** (cleaned, conformed, PII-handled), and **gold** (ML-ready, packaged, scored). Layers never skip forward; promotion requires explicit gates (§13).

P-04 · Reproducible builds

Any released dataset version **MUST** be byte-reconstructible from its lineage manifest, given access to the original sources and the pipeline code at the manifest's git SHA. No "I think we used this script" recovery paths.

P-05 · Privacy at ingest

P-07 · Modality-agnostic core, modality-specific edges

The orchestration, lineage, QA, license and audit layers do not know about modalities. Modality-specific tools (CV viewers, audio waveform labelers, geospatial tilers) plug in at well-defined interfaces.

P-08 · Async, idempotent, auditable

All non-trivial operations run as durable, idempotent jobs with retries, with structured audit events emitted to one log spine. No browser sessions own multi-hour work.

P-09 · Buyer-grade trust signals are visible

Provenance, license, PII status, freshness, schema, QA score and sample preview are exposed as first-class UI primitives, not buried in attachments. Trust is operational, not marketing.

P-10 · Operators before self-service

Internal admin workflows precede buyer/supplier self-service for the first 12 months. Self-service ships only after operator workflows for that capability are stable and load-bearing.

P-11 · Regulated by default

The platform is built to a posture compatible with EU AI Act Article 10, GDPR, CCPA/CPRA, and

PII detection runs at the bronze boundary. Sensitive data never reaches silver in the clear without a recorded justification, a recorded permitted-use, and a recorded retention policy.

P-06 · Licenses are code

Every record's permitted uses, derivative rights, exclusivity, geography, channel and term live in structured fields evaluated by policy code. PDF contracts attach as evidence; they are not the source of truth.

SOC 2 Type II readiness from day one. Markets without these obligations cost nothing extra; markets with them are accessible.

P-12 · Boring infrastructure

We choose mature, well-instrumented infrastructure with strong operational track records. Novelty in the dataset operations layer; conservatism everywhere else.

"Provenance, freshness, license, PII, QA score and sample preview are not metadata — they are the product."

Five planes. One data fabric.

The Caudals platform is structured as five horizontal planes, each with a clear responsibility and a clear interface to its neighbours. Anything new must fit somewhere on this map; if it does not, the map is wrong and must be updated before code lands.

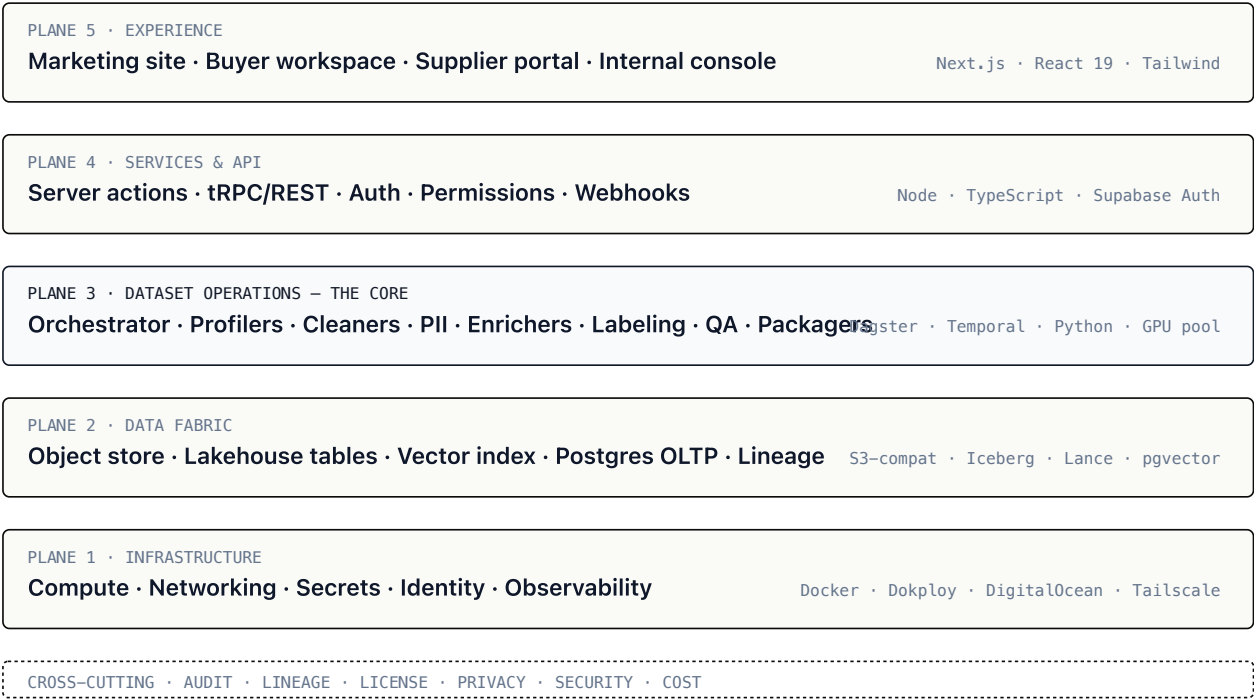


FIG 04.1 — FIVE-PLANE ARCHITECTURE; CROSS-CUTTING CONCERNS ARE PLATFORM-WIDE. FIG. 04.1

Plane responsibilities

PLANE	OWNS	MUST NOT
5 · Experience	Rendering, IA, locale, accessibility, design system, optimistic UI.	Hold business invariants. Talk to the data fabric directly.
4 · Services	Auth, RBAC, server actions, REST/tRPC routes, webhooks, idempotency, audit emit.	Run multi-minute work synchronously. Embed pipeline logic.

PLANE	OWNS	MUST NOT
3 · Operations core	Pipeline orchestration, modality workers, GPU/CPU job execution, lineage emission, run state.	Render UI. Hold customer-facing identity.
2 · Data fabric	OLTP records, lakehouse tables, object storage, vector indexes, lineage graph.	Run business logic. Make external calls.
1 · Infrastructure	Compute, networking, identity, secrets, deployment, monitoring.	Hold business state.

Deployment topology

EDGE / PUBLIC

Marketing site, blog, contact & waitlist APIs, public catalogue (when released).
Cached, rate-limited, isolated from operations VPC.

APP / PRIVATE

Internal console, buyer/supplier auth surfaces, server actions, webhooks.
Behind auth + Tailscale-restricted admin.

OPERATIONS VPC

Postgres, lakehouse, object store, orchestrator, GPU pool, vector index, observability stack. No public ingress beyond signed URLs and approved API gateway paths.

Boring where we can. Specialized where the moat lives.

The stack favours mature, instrumented technology with strong operational track records, and reserves novel choices for the dataset operations plane (§3) where Caudals' competitive moat is built. Every selection below either continues the existing repo direction or has an explicit upgrade path documented in §30.

Plane 5 · Experience

LAYER	CHOICE	WHY	ALTERNATIVES CONSIDERED
Framework	Next.js 16 App Router, React 19, TypeScript	Existing repo direction; SSR + RSC fit B2B latency profile.	Remix, SvelteKit
UI	Tailwind v4 + Radix UI + shadcn primitives	Editorial-grade tokens (§DESIGN.md) already in place.	Mantine, Chakra
Charts	Recharts + custom SVG, ECharts for ops dashboards	Lightweight, restrained, design-system-friendly.	Plotly, D3 raw
Tables	TanStack Table, virtualized rows for ops grids	Standard for ops-grade screens with 10k+ rows.	AG-Grid (license cost)
i18n	Existing translation pipeline; en + es as canonical	Spain locale is a strong market signal.	next-intl direct

Plane 4 · Services & API

LAYER	CHOICE	WHY	ALTERNATIVES
Runtime	Node 22 LTS, TypeScript strict mode	Continuity; ecosystem.	Bun (revisit at year 2)

LAYER	CHOICE	WHY	ALTERNATIVES
API style	Server Actions internally; tRPC for buyer/supplier; signed REST for partner integrations	Type-safe across boundaries; minimal contract drift.	GraphQL (overkill at this scale)
Auth	Supabase Auth + RLS; org-scoped roles; SCIM-ready for enterprise buyers	Already wired; RLS is the primary enforcement.	Auth0, WorkOS
Webhooks	Idempotent receivers, replay log, signed events	Required for Stripe + supplier-side integrations.	—
Email	Resend with audited templates & suppressions	Transactional + light marketing.	Postmark
Payments	Stripe (Connect for supplier revenue share)	Standard for B2B; revenue share via Connect transfers.	Adyen

Plane 3 · Dataset operations core

LAYER	CHOICE	WHY THIS, NOT THE OBVIOUS THING
Asset orchestrator	Dagster	Datasets-as-assets is the right mental model; software-defined assets, partitions, backfills, type-checked IO managers.
Durable workflows	Temporal	Multi-day human-in-the-loop labeling and supplier interaction flows must survive process death; activities + signals fit perfectly.
Compute	Python 3.12 workers; GPU pool on-demand for CV/audio/embeddings	Dagster ops dispatch to a GPU pool via Kubernetes or DO managed nodes.
Tabular engine	Polars + DuckDB (in-process), Spark only for >TB jobs	Polars is dramatically faster than pandas at our scale; DuckDB for SQL-shaped exploration.
Profiling	whylogs (streamed sketches) + ydata-profiling for one-shot reports	whylogs is privacy-safe and prod-runnable; ydata for human-readable reports.
Quality gates	Great Expectations (suites) + Pandera (typed dataframes)	GE for declarative assertions; Pandera as type-system in pipeline code.

LAYER	CHOICE	WHY THIS, NOT THE OBVIOUS THING
PII / privacy	Microsoft Presidio + spaCy NER + custom recognizers; OpenDP for tabular DP synthesis	Presidio's recognizer registry is extensible; OpenDP is the only mature DP toolkit.
Labeling	Label Studio embedded; CVAT for video; Argilla for LLM feedback	OSS, scriptable, multi-modal; we run them as managed components, not external services.
Active learning	FiftyOne Brain + Lightly-style embedding selection	Uncertainty + diversity sampling; pluggable behind a Caudals interface.
Synthetic data	SDV (multi-table) + Gretel for DP tabular when needed	SDV is open and composable; Gretel for regulated DP guarantees.
Document parsing	Unstructured + LlamaParse + AWS Textract fallback	Layout-aware OCR is the bottleneck for document datasets.
Embeddings	BGE / E5 family + custom domain encoders	OSS, on-prem-able, no vendor lock-in for vectors.

Plane 2 · Data fabric

LAYER	CHOICE	WHY
OLTP	Postgres 16 (Supabase self-hosted)	Existing infrastructure; pgvector, RLS, FDW headroom.
Object store	DigitalOcean Spaces (S3-compatible) + R2 for cross-region delivery	S3 API compatibility; existing repo wiring.
Lakehouse tables	Apache Iceberg on object store, queried by DuckDB / Trino	Vendor-neutral table format; works with every engine.
Tensor / unstructured	Lance format for image/video/audio collections; WebDataset for streaming training	Random-access columnar tensors; GPU-friendly streaming.
Vector index	pgvector for catalogue search; Qdrant for high-cardinality similarity at build time	pgvector co-locates with OLTP; Qdrant for filtered ANN at scale.
Versioning	lakeFS over object store	Git-like branches/commits over petabyte data without copies.

LAYER	CHOICE	WHY
Lineage	OpenLineage events → Marquez backend	Vendor-neutral spec; Dagster / Spark / Airflow all emit it.
Cache / queue	Redis (BullMQ) for low-latency jobs; Temporal for durable	Two-tier queue: ephemeral vs durable.

Plane 1 · Infrastructure & cross-cutting

LAYER	CHOICE	LAYER	CHOICE
Container runtime	Docker + Dokploy on DigitalOcean VPS (current)	Logs	OpenTelemetry → Grafana Loki
Orchestration (year 2)	Managed Kubernetes when GPU pool exceeds 4 nodes	Metrics	Prometheus + Grafana
Networking	Tailscale for admin, Traefik / Kong for public	Traces	OpenTelemetry → Tempo
Secrets	Doppler / Infisical; Supabase Vault for in-DB secrets	Errors	Sentry
CI/CD	GitHub Actions → Docker Hub → Dokploy	Analytics	Umami (existing) for marketing; PostHog for product

"Datasets are first-class assets. Pipelines produce assets; assets are the unit of release. Choose tools that respect that hierarchy."

PRINCIPLE P-02

PART II

The Dataset Operations Platform.

Thirteen sections describing the core engineering system — the pipeline that turns raw supplier data into ML-ready datasets across eight modalities. This is where the company's value is created. The rest of the platform exists to feed and harvest from this core.

06

Lifecycle &
medallion

07–15

Nine pipeline
stages

16–17

Lineage &
derivation

18

Multi-modality

From raw record to released asset, nine gates.

Every dataset built at Caudals passes through the same nine-stage pipeline, regardless of modality, customer, or source. The pipeline is expressed as Dagster software-defined assets composed by a build plan; each gate is a contract, not a step. Data does not advance to the next gate until the previous one has emitted a signed, lineage-recorded promotion event.

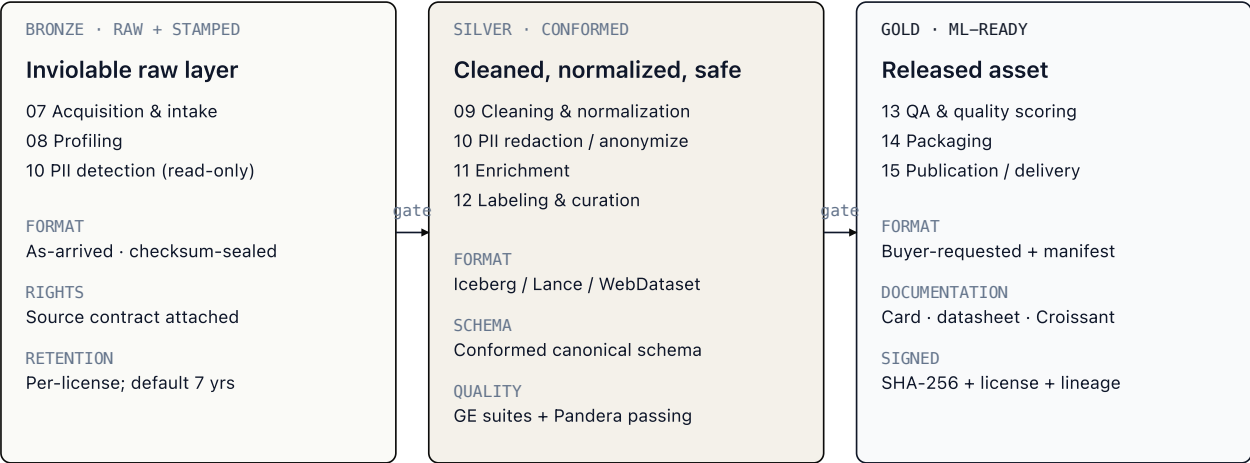


FIG 06.1 — BRONZE → SILVER → GOLD; GATES BETWEEN LAYERS ARE CONTRACTS.

FIG. 06.1

Why medallion, specifically

Medallion architecture maps cleanly onto Caudals' commercial model: bronze corresponds to *raw supplier asset*, silver to *build work-in-progress*, and gold to *SKU*. Promotion events between layers correspond to commercial events: bronze→silver triggers the point at which the build clock starts; silver→gold triggers the QA release that unlocks billing or revenue share.

The nine pipeline stages

#	STAGE	PURPOSE	ARTEFACTS PRODUCED
07	Acquisition & intake	Receive, fingerprint, register, and seal raw source material with provenance.	Bronze partition, source manifest

#	STAGE	PURPOSE	ARTEFACTS PRODUCED
08	Profiling	Inspect schema, distributions, types, missingness, duplicates, drift markers.	Profile report, sketches
09	Cleaning & normalization	Standardize values, units, encodings; deduplicate; resolve malformed records.	Silver partition (clean)
10	Privacy & PII handling	Detect, mask, anonymize, pseudonymize; document residual risk.	PII map, redaction log
11	Enrichment	Add derived features, joins, taxonomies, geo-coding, embeddings.	Silver-enriched partition
12	Labeling & curation	Annotate, review, resolve disagreement; sample for active learning.	Label batches, ontology
13	QA & quality scoring	Verify against acceptance criteria; produce scorecard and exception report.	QA report, score, gate verdict
14	Packaging	Convert to buyer-requested formats; produce dataset card, datasheet, Croissant manifest.	Gold artefact + manifests
15	Publication / delivery	Sign, license-stamp, deliver via signed URL, S3 share, API or warehouse share.	Delivery receipt, acceptance evidence

Sealing source material with provable provenance.

Acquisition is the first and most legally consequential stage. Every record entering the system is fingerprinted, attached to a contract, and sealed into the bronze layer with a provenance manifest. Once sealed, bronze data is read-only for the rest of the platform's lifetime.

Supported intake channels

Pull-mode

- **Object storage shares.** S3, R2, GCS, Azure Blob via short-lived credentials.
- **Database snapshots.** Postgres, MySQL, Snowflake, BigQuery, Redshift via read-only roles or share grants.
- **API connectors.** REST/GraphQL pulls with cursor-based pagination, retries, and replay.
- **Warehouse shares.** Snowflake Secure Data Sharing, BigQuery Authorized Datasets, Databricks Delta Sharing.
- **Public scrapers.** Whitelisted domains only, robots.txt-aware, rate-limited; for enrichment, never as primary source.

Push-mode

- **SFTP.** Per-supplier, key-only, isolated drop zones; events on drop.
- **Signed upload URLs.** Multi-part, resumable, tied to a supplier asset registration.
- **Email-to-bucket.** For small artefacts, with verified-sender allow-listing.
- **Physical media.** Encrypted drives mailed in; ingest at our facility under audit camera.
- **Webhook.** Supplier emits change events; Caudals pulls deltas.

The intake contract

Every intake event creates an immutable record:

```
// supplier_intake_event - bronze write
{
  "intake_id": "in_01H8ZG...",
  "supplier_asset_id": "sa_01H8ZG...",
  "channel": "sftp",
  "received_at": "2026-05-09T11:42:18Z",
  "received_by": "caudals-ingest-svc-prod-04",
  "contract_ref": "ct_01H8...#cl=4.2",    // license clause pinned
}
```

```

"jurisdiction": "EU-ES",
"object_uri": "s3://caudals-bronze/sa/01H8ZG/2026-05-09/raw.parquet",
"bytes": 2_847_113_402,
"sha256": "7c3b...b9e2",
"signed_by_supplier": true,
"caudals_signature": "ed25519:...",
"chain_of_custody": ["sftp-edge-03", "intake-svc-04", "bronze-store-eu-1"]
}

```

Required at intake (gate G-1)

- 01 **Provenance manifest.** Source organization, contract reference, jurisdiction, sender identity, transport channel, timestamp.
- 02 **Permitted-use declaration.** Structured: train, fine-tune, eval, infer commercial, infer non-commercial, redistribute. Per-record granularity if the contract requires it.
- 03 **Sensitivity flag.** One of: PUBLIC CONFIDENTIAL PII-PRESENT
PHI / PCI / SPECIAL-CATEGORY
- 04 **Refresh declaration.** One-shot, scheduled, on-event, or perpetual.
- 05 **Retention & right-to-erasure posture.** How DSARs propagate.
- 06 **Quarantine on suspicion.** Any of: malformed manifest, missing contract, unverifiable sender, unexpected sensitivity. Quarantine is a holding bucket with no downstream visibility.

Operational guarantees

ATOMICITY

Multi-part uploads commit only after manifest validation passes. No half-loaded partitions.

REPLAY

Every intake is reproducible from its event log. Failed intakes can be re-run without contacting the supplier.

AUDIT

Every read of bronze data is logged with caller identity, purpose, and downstream artefact ID.

Look at the data before you commit to a build.

Profiling produces a structured understanding of the source material: schema, types, distributions, cardinalities, missingness, duplication, suspected sensitive fields, and modality-specific signals (image resolution histograms, audio sample-rate distribution, document layout inventory). Every build plan is a function of its profile.

Profile dimensions, by modality

MODALITY	CORE PROFILE SIGNALS	MODALITY-SPECIFIC SIGNALS
Tabular	Row count, column types, null rates, cardinality, value distributions, key uniqueness, foreign-key suggestions.	Time-of-day skew, geographic skew, outlier clusters, duplicate keys.
Text	Token count, language mix, length distribution, dedup rate, encoding cleanliness.	Topic clusters, toxicity prevalence, contamination overlap with public benchmarks.
Image	Count, resolution histogram, aspect ratio, color depth, format inventory, near-duplicates via perceptual hash.	Object-class prior (zero-shot), exposure / blur quality, embedded metadata (EXIF/GPS).
Video	Duration histogram, fps, codec, resolution, scene-change density.	Subject coverage, audio-track presence, captions presence, redaction need.
Audio	Duration, sample rate, channels, codec, SNR estimate.	Speaker count estimate, language ID, voice activity ratio.
Geospatial	Bounding box, CRS, feature density, temporal extent.	Resolution inconsistency, edge artefacts, sensor mix.
Time series	Sample rate, gap distribution, regime changes, seasonality fingerprints.	Sensor drift, calibration jumps, blackout windows.

MODALITY	CORE PROFILE SIGNALS	MODALITY-SPECIFIC SIGNALS
Document	Page count, layout types (single/multi-column, tables, forms), language mix, OCR confidence.	Signature presence, tabular-data ratio, redaction artefacts.

Profile artefacts

Streamed sketches

whylogs profiles run inline with ingestion to produce privacy-preserving, mergeable sketches per partition. These are the production-runnable signal: they survive scale, are mergeable across time windows, and can be exposed safely without leaking row-level data.

Human-readable report

ydata-profiling produces a one-shot HTML report on a representative sample, attached to the build plan. This is the artefact the operator reads; the streamed sketches are what the platform reasons over.

Profile gate (G-2)

Promotion to the cleaning stage requires:

- **Schema fingerprint** registered in the schema registry (with diff against any prior version of this asset).
- **Suspected-PII column list** reviewed by an operator (the actual redaction happens in §10).
- **Encoding/format anomalies** resolved or quarantined.
- **Volume / completeness** within the supplier's declared envelope; any >15% deviation triggers review.
- **Suitability call** — operator records whether the data is suitable for the proposed buyer brief or supplier offer; if not, the asset returns to the supplier with feedback.

Profile-driven build planning

The profile output drives the build plan automatically: e.g., a column flagged 99.97% unique with a tax-ID-shaped pattern auto-routes to the PII recogniser pool; a column with 80% nulls auto-flags for either drop or imputation discussion; an image set with 40% near-duplicates auto-queues perceptual deduplication. The operator approves or overrides; nothing runs silently.

"A dataset's profile is not metadata. It is the only honest statement we can make about the data before we transform it."

Make the data say what it means, consistently.

Cleaning is the highest-leverage stage in terms of buyer-perceived quality. A buyer evaluating a sample notices encoding garbage, mixed units, inconsistent timestamps and broken joins long before they notice the labeling ontology. The cleaning stage is therefore over-invested relative to its lines-of-code share.

Cleaning operators — the toolbox

CLASS	OPERATOR	NOTES
Type repair	cast, parse-date, parse-numeric, parse-currency, infer-from-pattern	Rejects rows that fail with reason recorded.
Encoding	utf8-canonicalize, mojibake-fix, BOM-strip, line-ending-normalize	Common pre-2020 supplier exports are full of these.
Whitespace & case	trim, collapse, lower, title, sentence	Per-field; configurable.
Units	convert-currency, convert-temperature, convert-distance, normalize-time-zone	Authoritative reference tables versioned in the platform.
Identifiers	e164-phone, iso-country, iso-currency, iban-validate, vat-validate	Validation + canonicalization; rejection routed to repair queue.
Geo	geocode, reverse-geocode, h3-bin, country-from-coords	Cached against a managed local geocoder; rate-limited fallbacks.
Deduplication	exact-key, fuzzy-block, perceptual-hash, embedding-near-dup	Modality-aware; blocking + scoring + threshold.
Missing-value handling	drop, sentinel, group-mode, regression-impute, llm-fill (with caveat record)	Default is sentinel + flag; impute requires explicit operator approval.
Outliers	iqr-clip, zscore-clip, isolation-forest-flag	Default behaviour is to flag, not drop.

CLASS	OPERATOR	NOTES
Schema conformance	rename, project, reorder, type-coerce-strict	Output is the canonical Caudals schema for the dataset family.

Image / video / audio cleaning

Image	Video	Audio
Re-encode to canonical format (PNG/JPEG/WebP) at a set quality target. Strip EXIF except whitelisted fields. Resolve rotation tags. Flag corrupt frames. Perceptual deduplication via pHash + dHash.	Re-mux to canonical container (MP4 + H.264/H.265). Normalize fps to a target if labeling demands it. Extract keyframes for previews. Flag frames with embedded burnt-in PII (timestamps, names) for §10.	Resample to canonical rate. Normalize loudness (EBU R128). Strip ID3 tags. Voice-activity segment for downstream label efficiency. Flag DTMF / phone-number tones.

Cleaning gate (G-3)

- 01 **GE suite passes** against the silver schema with no critical failures.
- 02 **Pandera typed schema** validates without coercion warnings.
- 03 **Reject rate** within the build plan's tolerance, with rejected rows preserved in a side bucket for analysis.
- 04 **Sample diff** reviewed by operator: 50 random before/after pairs.
- 05 **Reproducibility check:** cleaning is deterministic given the same input + version-pinned operators; verified by hashing.

Cleaning is composable, not bespoke

Each operator is a Dagster op with a typed input/output. A build plan is a YAML manifest naming operators and their parameters. The platform ships a curated library of operators; teams add new ones via the `caudals/cleaning-operators` contribution repo with mandatory unit tests, deterministic-output tests, and a profiling benchmark.

PII is detected at ingest, not at delivery.

Privacy handling is the single most important non-functional requirement of the platform. A privacy incident is more damaging to the company than a year of missed sales targets. The architecture is designed so that PII presence is established before any silver-layer code touches the data, and so that the only PII that leaves the platform is PII a buyer's contract explicitly permits.

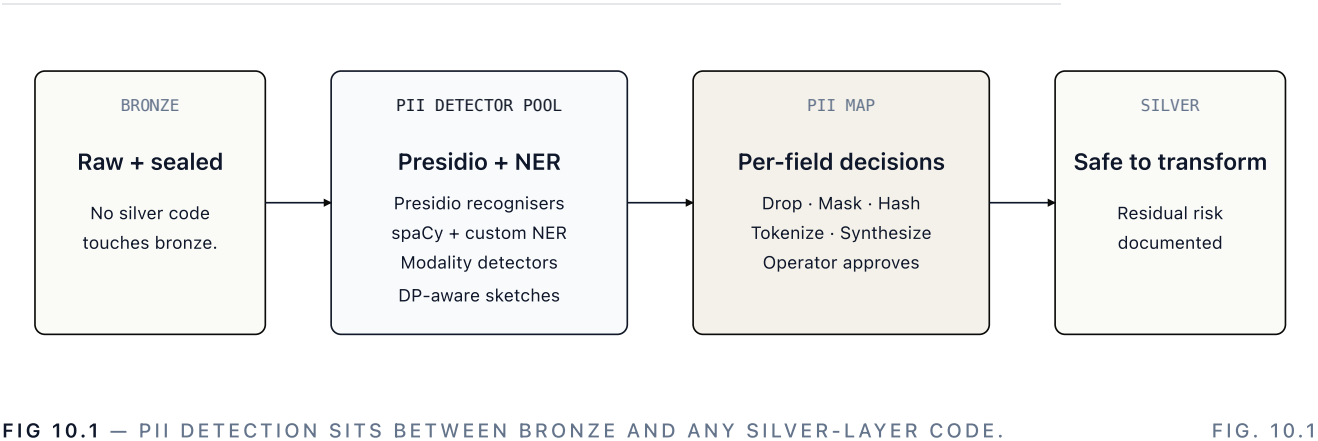


FIG 10.1 — PII DETECTION SITS BETWEEN BRONZE AND ANY SILVER-LAYER CODE.

FIG. 10.1

Detection

Detection is multi-layered:

- **Pattern recognisers** (Presidio): emails, phones, IBANs, passports, credit cards, SSNs, national IDs (per jurisdiction), MRNs, license plates.
- **NER models** (spaCy / transformers): person names, organizations, locations, with confidence scoring.
- **Custom recognisers**: per-supplier patterns (employee ID schemas, internal IDs that act as quasi-identifiers).
- **Modality detectors**: face detection (image/video), license-plate detection (image/video), voice biometrics (audio), embedded text in images (OCR + recogniser).
- **Quasi-identifier risk model**: flags column combinations whose joint distribution allows re-identification (k -anonymity < 5 over expected adversary's external data).

Treatment palette

TREATMENT	WHAT IT DOES	WHEN TO USE
Drop	Remove the field entirely.	Field offers no buyer value.

TREATMENT	WHAT IT DOES	WHEN TO USE
Mask	Replace with a sentinel ([REDACTED]).	Field exists for structure; content is sensitive.
Hash	Keyed cryptographic hash with a per-build pepper.	Joining is needed but identity is not.
Tokenize	Reversible token in a vault Caudals controls.	Reversibility is required for legal hold; never delivered.
Pseudonymize	Stable opaque ID per entity within the dataset.	Entity coherence required without identity.
Generalize	Bin to coarser bucket (age → band, ZIP → region).	Useful resolution preserved; reduces re-id risk.
Synthesize	Replace with statistically plausible synthetic values.	Tabular release where format must be preserved.
DP-noise	Differentially-private aggregates only.	Aggregate datasets for high-sensitivity domains.
Blur / pixelate	Image / video masking of faces, plates, screens.	CV datasets where context matters but identity does not.
Voice swap	Replace speaker identity while preserving prosody.	Audio datasets requiring speaker non-identifiability.

Privacy gate (G-4)

- 01 **PII map** reviewed and signed off by the operator and (for special categories) the privacy lead.
- 02 **Coverage report** proves treatment was applied to every flagged location.
- 03 **Residual-risk note** documents any acknowledged exposure and its commercial justification.
- 04 **Reverse map** (token vault, hash pepper) is stored separately, encrypted, with KMS-controlled access.
- 05 **DPIA reference** attached if special-category data is handled.

Add value carefully. Most records do not need it.

Enrichment is the most over-applied stage in industry data pipelines. Caudals enriches only when the buyer's brief or the catalogue strategy requires it, and never silently. Every enrichment must declare its source, its license, and the columns it adds; downstream consumers can opt out at delivery.

Enrichment classes

CLASS	EXAMPLES	LICENSE POSTURE
Reference data	Country / region taxonomies, ISO codes, currency conversion, calendar/holidays.	Caudals-licensed reference tables; permissively redistributable in derivatives.
Geospatial	Reverse geocoding, H3 indexing, administrative-boundary joins, weather context.	Mixed; weather/admin is OSM/governmental open; commercial sources flagged separately.
Categorical	Industry codes (NAICS/SIC/NACE), product taxonomies, MeSH/UMLS for medical.	Each taxonomy versioned and license-tracked.
Derived features	Statistical features (rolling means, deltas), time-bucketed summaries.	Inherits from underlying data.
Embeddings	Sentence / image / audio embeddings for downstream ML use.	Tracked with model identity, version, and any model-license restrictions.
LLM-derived	Summaries, classifications, structured extractions.	Treated as <i>derived</i> : the model used is recorded; its license propagates to the derivative.
External joins	Public records, sanctions, OFAC/PEP, third-party paid data.	External-data licenses tracked per row if needed.

LLM-as-enrichment, with caution

LLM-generated enrichments (summarization, structured extraction, classification) are useful but legally non-trivial. The platform records:

- **Model identity and version** — a hash, not a brand name.
- **Prompt template version** — tracked in a prompt registry.
- **Decoder settings** (temperature, max tokens) for reproducibility.
- **Confidence signal** if available (logprobs, self-rating).
- **Spot-check rate** — mandatory human sample of LLM output.
- **Cost ledger** per build for budget control.

Enrichment gate (G-5)

- 01 **Enrichment manifest** declares every added column with source, license, version, and computation method.
- 02 **Spot-check pass**: human review on a stratified sample, with disagreement rate below the build's threshold.
- 03 **Independence test**: enrichment columns are derivable purely from the silver layer; no information leaks from gold-only sources.
- 04 **License compatibility**: every enrichment's license is compatible with the dataset's intended buyer-side use.

"If the enrichment cannot be reproduced from a manifest, it is not an enrichment — it is folklore. Folklore does not ship."

Three strategies, one review fabric.

Caudals does not pick a labeling philosophy. It supports human-in-the-loop annotation, programmatic / weak-supervision labeling, and active learning in one fabric, because real production datasets need all three. The review and quality-control layer is shared; the production strategies differ.

Three production strategies

A · HUMAN IN THE LOOP

Annotation by trained reviewers

Multi-pass annotation in Label Studio / CVAT with reviewer queues, gold-question seeding, agreement scoring, adjudication. Used when ground truth requires expert judgement (medical, legal, regulated industries) or when programmatic strategies cannot meet the quality bar.

B · PROGRAMMATIC / WEAK

Labeling functions + label model

Snorkel-style: heuristics, regex, ontology-rule hits and small classifiers vote on each record; a label model resolves conflicts. Used for high-volume text and tabular labeling, often as a first pass to feed reviewer queues. LLM-as-labeler is implemented as a labeling function with its model/prompt logged.

C · ACTIVE LEARNING

Loop with model uncertainty

An assistant model is trained on early labels; its predictions inform sampling: high-uncertainty + diverse-embedding + boundary samples are routed to reviewers next. Reduces the labels needed to reach a quality target by 3–10x on most CV / NLP tasks at our scale.

Ontology & schema control

Each labeling project owns an **ontology**: the structured definition of classes, attributes, relations, and per-class instructions. The ontology is versioned in the platform; changing it during a build requires a documented migration of in-flight labels.

- **Class definitions.** Name, definition, positive/negative examples, edge-case notes.
- **Hierarchies.** Parent-child relations and inheritance.
- **Attributes.** Per-class attribute schemas (e.g., bounding box + occlusion + truncation flags).
- **Relations.** Cross-instance relations (e.g., "is part of", "located at").
- **Conflict rules.** Which class wins when annotators disagree.

Reviewer queue & quality control

MECHANISM	WHAT IT DOES
Gold questions	Pre-labeled items inserted into reviewer queues; deviation triggers review of the reviewer.
Multi-pass with adjudication	Two or three independent labels per item; disagreements routed to a senior adjudicator.
Agreement metrics	Cohen's kappa / Krippendorff's alpha / IoU per class; tracked over time.
Confidence-aware routing	Low-confidence items get more reviewers; high-confidence items skip review when programmatic source agrees with one human pass.
Reviewer cohorts	Each reviewer has skill profiles; queue routes by skill match.
Time tracking	Per-item time recorded; outliers (too fast, too slow) sampled for review.
Cleanlab pass	Confident-learning scan finds suspected label errors after first pass; requeues them.

Workforce composition

Three reviewer pools are supported:

- **In-house specialists.** Caudals staff handling sensitive / regulated / expert-judgement work.
- **Vetted partner workforce.** Contracted reviewers via partner labeling firms with NDAs and quality SLAs; managed entirely through Caudals' own UI to prevent data sprawl.
- **Subject-matter experts.** Doctors, lawyers, mechanics, engineers contracted per-project for adjudication.

Data isolation

Data never leaves Caudals' VPC. Reviewers connect to a hardened web UI; download is disabled, screen-region restriction enforced for sensitive datasets, watermarking applied, and review activity is fully audit-logged. Special-category data is reviewed only by background-checked staff in a policy-restricted environment (no clipboard, no screenshot, no outbound network).

Labeling workflow — the canonical flow

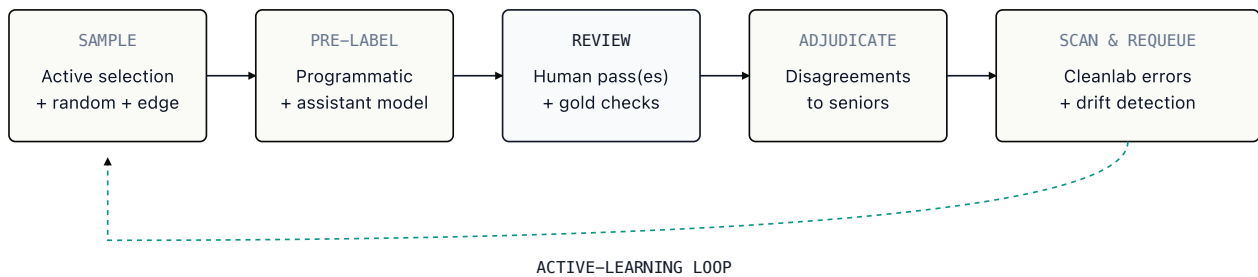


FIG 12.1 — LABELING FABRIC: PRE-LABEL, REVIEW, ADJUDICATE, SCAN, LOOP.

FIG. 12.1

Labeling gate (G-6)

- 01 **Coverage:** every silver record either labeled, deferred with reason, or excluded with reason.
- 02 **Inter-annotator agreement:** Krippendorff's alpha $\geq$ threshold per class; failing classes block release.
- 03 **Gold accuracy:** reviewer accuracy on gold questions $\geq$ threshold; below threshold returns to retraining.
- 04 **Cleanlab error rate:** estimated label-error rate $\leq$ threshold after final pass.
- 05 **Class distribution:** matches the build plan's intended distribution within tolerance, or rebalancing is documented.

Review UI requirements

- Modality-aware widgets (bounding box, polygon, mask, keypoint, audio segment, text span, document field).
- Side-by-side compare with prior pass; reviewer history visible.
- Keyboard-first interaction: a senior reviewer should never reach for a mouse.
- Hot-keys per ontology class are configurable.
- Inline ontology reference; no separate document.
- Disagreement view: visual overlay of all annotators on the same item.
- Confidence overlays from the assistant model.
- Audit trail per item: who, when, how long, what changed.
- Bulk actions with confirmation and reversibility.
- Network-loss-tolerant: reviewer never loses work to a flaky connection.

A scorecard the buyer trusts before they pay.

The QA stage produces three artefacts: an accept/reject verdict against the build plan's acceptance criteria, a public-facing quality scorecard that the buyer sees at sample preview, and a structured exception report used internally to decide release vs. rework.

Quality dimensions

DIMENSION	DEFINITION	MEASURED BY
Completeness	Required fields present and non-null at expected rates.	GE suite, profile delta vs. plan.
Validity	Values match declared types, ranges, and reference vocabularies.	Pandera schema, validator suites.
Consistency	Cross-field relationships hold (e.g., country in address matches phone).	Custom rules, GE compound expectations.
Uniqueness	Declared keys are unique; near-duplicates are below tolerance.	Hash + perceptual / embedding scans.
Timeliness	Records are within the freshness window declared on the listing.	Date column statistics, drift detector.
Accuracy	Labels match held-out gold within tolerance.	Gold-set evaluation, agreement metrics.
Representativeness	Distribution matches the declared population (geography, time, class balance).	Stratified statistics vs. plan.
Privacy	No PII present beyond the declared posture; reverse-id risk < threshold.	PII scanner re-run on gold; k-anonymity check.
Provenance	Every record traces to a permitted-use clause.	Lineage join over license vault.

DIMENSION	DEFINITION	MEASURED BY
Reproducibility	Build can be re-run from manifest with identical hash.	Replay test in CI before release.

The composite quality score

Each dimension produces a normalized score in `[0,1]`. The composite Q-score is a weighted geometric mean (so a single zero-rated dimension cannot be hidden by strong others). Weights are set per dataset family and locked at build start. The Q-score is exposed publicly on the catalogue listing alongside its component breakdown.

```
// quality_scorecard sample
{
  "dataset_version_id": "dv_01H8ZG...",
  "composite": 0.91,
  "dimensions": {
    "completeness": 0.99,
    "validity": 0.98,
    "consistency": 0.94,
    "uniqueness": 0.97,
    "timeliness": 0.88,
    "accuracy": 0.86,    // label-vs-gold
    "representativeness": 0.82,
    "privacy": 1.00,
    "provenance": 1.00,
    "reproducibility": 1.00
  },
  "verdict": "release",
  "exceptions": [
    {"code": "REPR_GEO_SKEW", "detail": "DE under-represented vs plan by 7%"}
  ]
}
```

Bias & fairness checks

For datasets feeding regulated AI systems, an additional bias panel runs: representational balance across declared protected attributes (where present and lawful), label distribution fairness, and (where the buyer brief calls for it) a downstream-model bias probe trained on a fraction of the data.

Acceptance evidence

Every release produces an *acceptance evidence* bundle: the scorecard, the exception report, the lineage manifest, the licensing summary, and a sample preview. This bundle is what the buyer signs against on delivery and what auditors look at later. It is immutable.

One canonical asset, many packaged forms.

A released dataset has a single canonical representation in the lakehouse and a set of packaged deliverables generated on demand for the formats and frameworks each buyer requires. Packaging is a deterministic transformation of the canonical asset; it never alters semantics, only encoding.

Canonical representation

For each modality, Caudals chooses one canonical lakehouse format:

MODALITY	CANONICAL	WHY
Tabular	Iceberg + Parquet (zstd)	Ubiquitous, columnar, partition-pruned, vendor-neutral.
Text	Parquet (text + metadata) + JSONL mirror	Parquet for engines; JSONL for tokeniser pipelines.
Image	Lance dataset + media references	Random access for training; references resolve to S3 objects.
Video	Lance index over MP4 chunks + frame manifests	Streamable in training pipelines.
Audio	Lance index over WAV/FLAC + segment manifests	Per-segment labels join cleanly.
Geospatial	STAC catalog + GeoParquet + Cloud-Optimised GeoTIFF	Industry-standard interchange.
Time series	Iceberg / Parquet partitioned by time + entity	Time travel and partition pruning.
Document	Parquet (page-level structured) + original-PDF references	Both layout-aware features and original retained.

Packaging targets — the export matrix

TARGET	USE	NOTES
Parquet (zip)	Universal tabular delivery.	Default for warehouse / Spark / Polars consumers.
CSV	Legacy / spreadsheet consumers.	Strict UTF-8, RFC-4180; per-column type sidecar.
JSON / JSONL	NLP and document pipelines.	One record per line; canonical key order.
COCO JSON	CV detection / segmentation.	Image references + annotations; ontology-mapped.
YOLO txt	YOLO-family training.	Class index file + per-image annotations.
TfRecord	TensorFlow training.	Sharded; manifest of shards.
WebDataset	PyTorch streaming.	Sharded tar; lossless mapping from canonical Lance.
HF Datasets	Hugging Face consumers.	Includes Croissant manifest.
Delta Sharing / Snowflake	Warehouse-native delivery.	For buyers who never want to download.
REST / gRPC API	Streaming consumption.	Per-record paged with cursors.

Required documentation, packaged with every release

- 01 Dataset card.** Hugging Face style: purpose, sources, structure, splits, considerations, limitations.
- 02 Datasheet.** Gebru et al. style: motivation, composition, collection, preprocessing, uses.
- 03 Croissant manifest.** Machine-readable JSON-LD for ML toolchains.
- 04 Schema & data dictionary.** Per-column types, descriptions, value ranges, references.
- 05 Quality scorecard.** The artefact from §13.
- 06 Lineage / provenance summary.** Source, contracts, transformations, tool versions, run hashes.
- 07 License & permitted-use summary.** Plain-language plus structured permissions.
- 08 Privacy summary.** PII handling posture, residual risk, DSAR procedure.

09 **Refresh policy.** Cadence, expected drift, deprecation policy.

10 **Sample preview.** 100 representative records or 30 representative items per modality.

Packaging gate (G-7)

Every packaged artefact must:

- Round-trip back to the canonical with identical content hashes (lossless transformation).
- Carry a `manifest.json` at the package root with version, build hash, signing key, and embedded Croissant.
- Include all ten required documents above.
- Pass an automated consumer test that loads the package in its native framework (e.g., `datasets.load_dataset(...)` for HF) and validates record count + schema.

Hand-off the buyer can verify, audit later.

Delivery is the moment the platform's promises become contractual. Caudals' delivery posture treats the buyer's first 24 hours with the artefact as the most important hour of the engagement: the artefact must load, the documentation must be coherent, and the acceptance evidence must be machine-verifiable.

Delivery channels

CHANNEL	USE	NOTES
Signed download URL	Default for one-shot delivery.	Time-limited; per-buyer; multi-part for > 5 GB; resumable.
Buyer-side S3 bucket	Buyer provides write target.	Cross-account role-assumption; encrypted in transit.
Caudals S3 read share	Buyer reads from Caudals' bucket.	Per-buyer prefix; access keys rotated per delivery.
Snowflake / Delta share	Warehouse-native consumption.	Provider-managed; per-share license enforced.
REST / gRPC API	Streaming or query-based.	Rate-limited per contract tier.
Physical encrypted media	Air-gapped buyers.	Logged, signed, returned for destruction.
Hugging Face mirror	Public catalogue listings only.	Caudals-managed namespace; license-stamped.

Catalogue listing vs. private offer

Catalogue listing

Public (or buyer-base-only) listing in the marketplace catalogue. Standard licensing tiers, public quality scorecard, sample preview, list

Private offer

Single-buyer or small-group offer. Customized license, exclusivity window, custom delivery

price, refresh cadence. Future surface; ships after the catalogue schema and listing UI are released (see §30).

cadence, NDA-gated sample preview. Default delivery mode through year one.

Acceptance & receipt

- 01 **Pre-delivery checksum.** Caudals signs the package; buyer verifies signature with Caudals' published key.
- 02 **Delivery event.** Buyer download / share grant emits a delivery event; the receipt is timestamped and hashed.
- 03 **Acceptance window.** Buyer has a contracted window (typically 7 days) to raise issues against the QA scorecard.
- 04 **Acceptance signature.** Buyer countersigns the receipt; the asset is now *accepted*; commercial events trigger.
- 05 **Audit retention.** Receipt + scorecard + lineage manifest are retained per the contract's audit term (default 7 years).

Refresh & subscription delivery

Subscription deliveries (recurring data feeds) ship as *increments*: each refresh produces a new dataset version with a delta manifest pointing to the previous version. Buyers consume either the latest version or a rolling window. Each increment carries its own quality scorecard and privacy verification — rights are not assumed to be inherited.

Right-to-be-forgotten propagation

When a supplier or upstream subject exercises a deletion right, the platform must propagate deletion to all derivative dataset versions still under retention. The propagation runs as a tracked job: every affected dataset version is rebuilt or marked *tombstoned*; buyers consuming subscription deliveries receive a deletion-required notice with an actionable manifest of affected records. This is a hard requirement of operating in EU / CA / regulated jurisdictions.

Every dataset version knows where it came from.

Lineage is the platform's connective tissue. It is what allows Caudals to defend dataset provenance to a buyer's procurement team, comply with a supplier's revocation, propagate a deletion request through derivatives, and rebuild any historical asset on demand. The lineage layer is non-optional and runs platform-wide.

Three nested versioning systems

SYSTEM	WHAT IT VERSIONS	GRANULARITY
Git	Pipeline code, build plans, ontologies, prompts, configs.	Per commit; hash is the identity.
lakeFS	Bronze, silver, gold partitions on object store.	Per commit on per-asset branches; petabyte-safe.
Iceberg / Lance time-travel	Table-level snapshots within a lakeFS commit.	Per snapshot id; cheap point-in-time queries.

The lineage event spec

Every transformation emits an OpenLineage-compliant event:

```
// OpenLineage RunEvent (abbreviated)
{
  "eventType": "COMPLETE",
  "eventTime": "2026-05-09T13:42:18Z",
  "run": { "runId": "r_01H8ZG...", "facets": { "caudals_build": { "buildId": "bd_01H8..." } } },
  "job": { "namespace": "caudals.ops", "name": "clean_normalize.v3",
    "facets": { "sourceCode": { "gitSha": "a8f3..." } } },
  "inputs": [ { "namespace": "caudals.bronze", "name": "sa/01H8ZG/2026-05-09",
    "facets": { "version": { "datasetVersion": "lakeFS:b3c1@8a91" } } } ],
  "outputs": [ { "namespace": "caudals.silver", "name": "ds/01H8ZG/clean",
    "facets": { "schema": { "fields": [ ... ] },
      "dataQualityAssertions": [ ... ] },
  ] }
```

```
} "caudals_license":{ "contractRef":"ct_01H8...#cl=4.2" } } }
```

The lineage graph

Events flow into Marquez (or an equivalent backend), producing a navigable graph of:

- **Datasets** — bronze, silver, gold partitions across versions.
- **Jobs** — pipeline operators with code SHA and parameters.
- **Runs** — specific executions with start/stop, status, metrics, logs.
- **Facets** — schema, quality assertions, license clauses, PII map references, model identifiers.

Provenance proofs

- 01 **Build manifest.** A YAML at the root of every gold release naming sources, transforms, parameters, code SHAs, model versions, and producing hash.
- 02 **Replay test.** CI re-runs a representative sample of the manifest in a clean environment and asserts hash equality.
- 03 **Sign-off.** Operators sign release manifests; signatures are stored alongside the asset.
- 04 **External attestation** (optional, regulated buyers): notarised hashes published to a transparency log.
- 05 **C2PA stamp** (image/video/audio): content credentials embedded for AI-trained media artefacts.

Time-travel queries

Operators and auditors can query *"what did this dataset look like on date X"* via Iceberg / Lance snapshot reads, joined to the lineage graph. The platform exposes a `caudals lineage trace <asset_id>` CLI and a UI surface in the admin console.

Datasets compose. The catalogue is its own raw material.

A core capability of Caudals is building new datasets from existing ones — combining, slicing, filtering, augmenting, mixing, or re-labelling assets the platform already owns. The derivation engine treats this as a first-class build, with a stronger licensing engine because every input asset's permitted-uses must compose into the new output's permitted-uses.

Derivation operators

CLASS	OPERATOR	NOTES
Slice	filter, stratify, sample, time-window, geo-window	Always recorded; never silent.
Combine	union, join, multimodal-pair	Schema reconciliation is explicit, not inferred.
Re-label	ontology-remap, hierarchical-collapse, expert-relabel	Original labels preserved as a hidden facet.
Augment	augmentation pipeline (image/audio/text), counterfactual generation	Augmentations marked; downstream consumers can opt out.
Synthesize	SDV / Gretel synthetic from real, with utility-vs-privacy report	Synthetic provenance is explicit; never blended with real silently.
Decontaminate	n-gram + embedding overlap removal vs. eval benches.	Required for LLM-training derivatives.
Curate	quality-score gating, deduplication, hard-negative mining.	Drives buyer-specific evaluation sets.

License composition algebra

Each input's license is a structured object with permissions and restrictions (training, fine-tuning, eval, commercial inference, redistribution, exclusivity, geographic, term, share-alike). The derivation engine

computes the intersection of these permissions for the output. If the intersection is empty for the buyer's intended use, the build is blocked at planning time, not at delivery.

```
// composed permitted-use computed at plan time
inputs: [
  ds_A.permits = {train:true, commercial_inference:true, redistribute:false, geo:"WW"},
  ds_B.permits = {train:true, commercial_inference:false, redistribute:false, geo:"EU"}
]
output.permits = intersect(inputs.permits)
                  = {train:true, commercial_inference:false, redistribute:false, geo:"EU"}

// buyer brief asks for commercial-inference + WW – build PLAN BLOCKED
```

Common derivation recipes

- **Domain corpus:** union of N text sources, deduplicated, decontaminated against eval benches.
- **Eval set:** stratified sample from a held-out partition, expert re-labelled, never trained on.
- **Hard-negative pool:** assistant-model errors mined and re-routed for review.
- **Multilingual variant:** same content in additional languages via translation pipeline + human review.
- **Synthetic-augmented training:** real seed + synthetic augmentation in a documented mix ratio.
- **Vertical slice:** single industry / geography / time-window cut from a broader catalogue asset.
- **Multi-modal pair:** image+caption, audio+transcript, document+structured-fields built from aligned sources.

Reusable build templates

Recurring derivations are saved as *build templates*: parameterised YAML manifests that operators can instantiate per buyer brief. Templates make pricing predictable and execution fast. The first templates Caudals ships are: `eval-set-build`, `domain-corpus-build`, `vertical-slice-build`, `multi-supplier-merge`, `synthetic-augmented-build`.

One platform. Eight ways to be a dataset.

The orchestration, lineage, license and audit layers are modality-agnostic (Principle P-07). Each modality plugs in via a contract that defines its canonical representation, profile signals, cleaning operators, privacy treatments, labeling widgets, QA dimensions and packaging targets. New modalities can be added by implementing this contract; nothing in the core changes.

01 Tabular

Use cases. Predictive modelling, fraud, churn, logistics, finance. **Canonical.** Iceberg+Parquet. **Tooling.** Polars / DuckDB / Spark. **Special concerns.** Foreign-key consistency, time-leakage controls, quasi-identifier risk. **Packaging.** Parquet, CSV, Snowflake share, REST API.

02 Text

Use cases. LLM pre-training, fine-tuning, RAG corpora, classification, NER. **Canonical.** Parquet+JSONL mirror. **Tooling.** spaCy, HF Datasets, Argilla, Tiktoker. **Special concerns.** Language detection, contamination vs. eval benches, toxicity audit, copyright posture. **Packaging.** JSONL, HF Datasets, parquet shards.

03 Image

Use cases. Detection, segmentation, classification, generative training, OCR, retrieval. **Canonical.** Lance with media references. **Tooling.** FiftyOne, Albumentations, Roboflow exports. **Special concerns.** EXIF/GPS stripping, face/plate redaction, perceptual deduplication, copyright/style inheritance for generative training. **Packaging.** COCO, YOLO, Lance, WebDataset.

04 Video

05 Audio

Use cases. ASR, speaker ID, audio events, music labelling. **Canonical.** WAV/FLAC + Lance index + segment manifests. **Tooling.** Label Studio audio, pyannote, Whisper for pre-label. **Special concerns.** Voice biometrics as PII, speaker consent, prosody preservation under anonymisation. **Packaging.** WAV/FLAC + JSONL labels, HF Audio.

06 Geospatial

Use cases. Climate, agriculture, logistics, urban analytics, defence. **Canonical.** STAC + GeoParquet + Cloud-Optimised GeoTIFF. **Tooling.** GeoPandas, TorchGeo, Kepler.gl. **Special concerns.** CRS reconciliation, sensor calibration drift, sovereign data restrictions, embargoes by jurisdiction. **Packaging.** STAC catalogue, GeoParquet, COG, MVT tiles.

07 Time series

Use cases. Forecasting, anomaly detection, industrial, finance. **Canonical.** Iceberg / Parquet partitioned by time + entity. **Tooling.** TimescaleDB, tsfresh, sktime. **Special concerns.** Sample-rate alignment, gap policies, seasonality fingerprinting, sensor drift, proper train/val/test temporal splits. **Packaging.** Parquet partitioned, Arrow IPC, REST cursor API.

Use cases. Action recognition, surveillance, autonomous systems, generative video.
Canonical. Lance index over MP4 chunks.
Tooling. CVAT, Encord-style temporal review, FFmpeg. **Special concerns.** Temporal dedup, shot-change-aware sampling, burnt-in PII, ID3/audio metadata, keyframe vs. dense annotation policies. **Packaging.** Per-clip MP4 + per-frame manifest, COCO video.

08

Document

Use cases. KYC, contracts, invoices, medical records, claims, due diligence. **Canonical.** Parquet (page-level structured) + original PDF references. **Tooling.** Unstructured, LlamaParse, Textract, layout-LM family. **Special concerns.** Layout fidelity, hand-written vs. printed, signature redaction, multi-page coherence, language mix. **Packaging.** Parquet+PDF bundles, JSONL, REST query API.

Cross-modal datasets

Many of the highest-value datasets are cross-modal: image + caption, document + structured fields, video + transcript, sensor + outcome. Cross-modal datasets are modeled as *aligned partitions* with a shared key column and modality-specific physical layouts. The platform exposes a `multimodal-pair` derivation operator that produces aligned packages.

PART III

Product surfaces.

Four sections describing how the platform is exposed to the four populations who interact with it: operators, buyers, suppliers, and the public. Operators come first; everything else descends from the operations console.

19	20	21	22
Operator console	Buyer workspace	Supplier portal	Public funnel

The operations control room.

The internal console is the operator-facing surface and the most feature-dense product the platform ships in year one. It exposes every record needed to qualify, build, audit, ship, and bill datasets. It is information-dense by design and built for keyboard-first power users.

Console module map

MODULE	WHAT OPERATORS DO HERE	ANCHOR RECORDS
Pipeline / Home	Triage queue across all sides; today's blockers; overdue gates; release calendar.	Cross-domain index.
Leads & Opportunities	Inbound buyer/supplier pipeline; qualification; scoping; stage transitions.	buyer_opportunity, supplier_opportunity.
Suppliers	Organizations, contacts, contracts, asset registry, rights vault.	supplier_organization, supplier_asset.
Buyers	Organizations, contacts, briefs, sample-review history, contracts, deliveries.	buyer_organization, dataset_brief.
Builds	Active builds; gate status; QA readouts; rebuild & replay tools.	build, run, gate.
Datasets	Catalogue of dataset families and their versions; lineage browser; sample preview.	dataset, dataset_version.
Labeling	Project list; reviewer roster; queue depths; agreement metrics.	labeling_project, label_batch.
Quality	QA reports, scorecards, exception log, bias panels.	qa_report.
Privacy & Rights	License vault, consent registry, DSAR queue, retention schedules.	license_clause, consent_record.
Catalogue & Offers	Listing copy, pricing, refresh cadence, exclusivity windows, sample-preview gating.	catalogue_listing, private_offer.

MODULE	WHAT OPERATORS DO HERE	ANCHOR RECORDS
Commercials	Quotes, contracts, invoices, supplier payouts via Stripe Connect.	quote, contract, invoice, payout.
Operations	Job queue, GPU utilisation, cost ledger, alerts, runbooks.	run, alert.
Audit	Cross-domain audit log search; export for regulators / buyers.	audit_event.
Settings	Roles, permissions, integrations, signing keys, locale, environment toggles.	operator_role, integration.

Power-user behaviours

- **Command palette** (⌘K): jump to any record by ID, asset name, or contract number; trigger common actions.
- **Saved views**: each module exposes named filters owned by the operator or shared with the team.
- **Bulk actions**: with explicit row-count confirmation; reversible where practical.
- **Inline editing** with optimistic UI and conflict detection.
- **Keyboard navigation** across grids; one-character shortcuts for stage transitions.
- **Audit overlays**: every record exposes who/when/what for any change, inline.

Layout principles (DESIGN.md)

The console follows the existing Caudals design system: white panels on light gray canvas, subtle borders, restrained shadows, emerald accent for actionable states, monospace for technical identifiers. Density is high but readable; the design must remain comprehensible under time pressure during release windows or supplier escalations.

Reference layout — "Build" detail

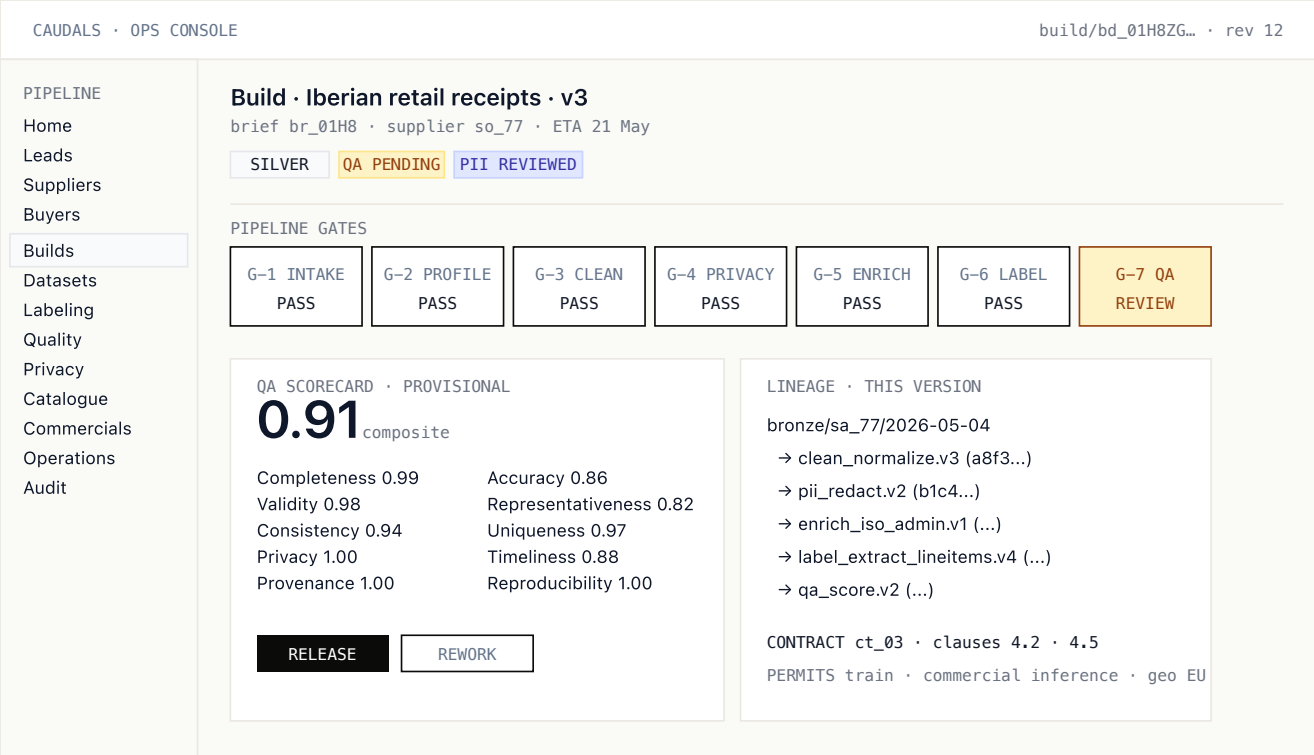


FIG 19.1 — REFERENCE LAYOUT FOR THE BUILD DETAIL SCREEN.

FIG. 19.1

A surface buyers actually want to log into.

The buyer workspace ships after operator workflows are reliable. It is the AI / ML team's window into their account: their datasets, their deliveries, their quality reports, their licenses, their refresh pipelines, and their support history. It must feel native to a buyer's other infrastructure tools (Snowflake, Databricks, MLflow), not like a marketplace storefront.

Module map

MODULE	WHAT BUYERS DO HERE
Catalogue browse	Filter the public / partner-only catalogue by modality, industry, geography, freshness, license posture, price band.
Dataset detail	Sample preview, schema, scorecard, datasheet, license summary, provenance summary, refresh policy, sample loader code (Python / SQL).
Brief intake	Submit a custom build brief: use case, modality, schema needs, volume, freshness, geography, sensitivity, target formats, target frameworks, budget, deadline.
Builds & deliveries	Per-engagement view: status, gate progress, sample previews, scorecards, quotes, contracts, deliveries, acceptance windows.
Subscriptions	Active recurring deliveries; refresh history; deltas; upcoming refreshes; pause / resume.
Licenses	Permitted-use summary across all owned datasets; geographic and channel constraints; expirations.
Integrations	S3 cross-account roles, Snowflake / Databricks share targets, signed-URL preferences, webhook receivers.
Team & access	Org members, roles, SSO, SCIM, audit log of buyer-side actions.
Billing	Invoices, payment methods, spend by engagement, budget alerts.
Support & messages	Threaded conversations with the assigned Caudals operator on each engagement.

Trust primitives in the buyer UI

Every dataset surface a buyer sees displays:

- **Quality scorecard chip** with composite score and dimension breakdown on hover.
- **License chip** with permitted-use icons (train, fine-tune, eval, commercial inference, redistribute) and geography.
- **Privacy chip** indicating PII posture (none, masked, anonymised with residual-risk note).
- **Provenance chip** with source-organization count and contract status.
- **Freshness chip** with last-refresh date and cadence.

Sample preview behaviour

Sample previews are watermarked, log every view, and are gated on a per-buyer NDA acknowledgment for non-public catalogue listings. Buyers can request access to a fuller preview through the workspace; operators approve or deny in the console.

Code snippets, on every dataset page

Each dataset page exposes ready-to-run code snippets for: Python (Polars / Pandas / HF Datasets), SQL (Snowflake / Databricks share), PyTorch (WebDataset), and a CLI download command. Buyers should be one click away from a working pipeline.

A side built for non-data-vendors.

Most suppliers are not data vendors. They are operationally rich companies (logistics, retail, healthcare, agriculture, manufacturing) who own valuable data but have never sold it. The supplier portal must feel low-friction, legally reassuring, and operationally simple. It is not a self-serve listing publisher; it is a managed onboarding surface.

Module map

MODULE	WHAT SUPPLIERS DO HERE
Onboarding	Org profile, KYC, signing authority, primary contact, jurisdiction.
Asset registry	Declare data assets: name, modality, volume, schema sketch, refresh, sensitivity, intended availability (private / catalogue).
Rights & consent	Provide and version the legal posture per asset: ownership, end-user consent, third-party content, AI-training rights, derivative rights, geography, exclusivity tolerance.
Sample upload	Submit a representative sample for Caudals' feasibility pass; signed upload URLs, multi-part, resumable.
Build participation	Track Caudals' progress on each engagement against this supplier; respond to clarification requests; approve schema decisions; sign off on the released dataset version.
Catalogue presence	(when activated) approve or block individual catalogue listings derived from supplier assets.
Revenue share	Earnings, payouts, contract terms; Stripe Connect-backed.
Refresh program	For recurring deliveries: schedule, change-management, deprecation notice.
Compliance & DSARs	Process incoming subject-rights requests; trigger propagation into Caudals' derivative datasets.

Onboarding flow

- 01 **Expression of interest.** Public contact form / direct sales.
- 02 **Qualification call.** Caudals operator assesses fit; opens a `supplier_organization` record.
- 03 **NDA & intake agreement.** Boilerplate but jurisdiction-aware; signed in-portal via e-sign.
- 04 **Asset declaration.** Supplier declares one or more candidate assets with structured rights metadata.
- 05 **Sample submission.** A small representative sample is uploaded and ingested under feasibility licence.
- 06 **Feasibility memo.** Caudals returns a memo with target schema, risk register, build plan, cost range, and go/no-go.
- 07 **Pilot or full build.** If green, Caudals scopes a pilot (typically) before a complete build, with the rights and revenue-share terms locked.

What suppliers should never have to do

- Build their own anonymisation pipeline.
- Negotiate AI-training-license language with each buyer.
- Operate buyer support, payments, or refund handling.
- Maintain dataset packaging in COCO / Parquet / TFRecord.
- Handle DSARs that span derivative datasets they did not produce.

All of the above is owned by Caudals. The supplier's job is to bring the data and the rights; the platform's job is to monetize them.

The front door — fast, credible, easy to contact.

The public surface is already in production: landing, contact, blog, waitlist, analytics. Its job is to convert latent demand into qualified conversations and to communicate a serious B2B data-operations posture to procurement teams researching the company. It is intentionally narrow until self-service surfaces are ready.

Public surface inventory

ROUTE	PURPOSE	STATUS
/	Landing — thesis, capabilities, sample case studies, hero waitlist.	Live
/contact	Buyer / supplier / partner contact intake.	Live
/blog	Editorial publishing for SEO and credibility.	Live
/blog/[slug]	Article pages.	Live
/api/contact	Server-side contact submission.	Live
/api/waitlist	Waitlist registration.	Live
/catalogue	Public catalogue (curated listings).	FUTURE — SHIPS IN ROADMAP M3
/customers	Trust page (logos, case studies, certifications).	FUTURE — SHIPS WHEN CONSENTED
/security	Public security & compliance posture page.	FUTURE — SHIPS AT SOC 2 READINESS
/legal/*	Privacy, ToS, DPA, supplier & buyer agreements.	Live (rolling updates)

Funnel instrumentation

- **Server-side capture** for contact and waitlist submissions, idempotent and audit-logged.

- **Lead enrichment** on submission: company match (employees, industry, region), routing to the appropriate operator.
- **Marketing analytics** via Umami (existing), product analytics later via PostHog when authenticated surfaces ship.
- **Spam & abuse controls:** rate limits, bot detection, jurisdiction-aware fields.

Editorial standard

The blog is treated as a serious editorial channel: every post owns a single argument, sourced and dated. Topics that move the buyer's procurement conversation forward (provenance, EU AI Act, dataset documentation, modality-specific quality) are prioritised over generic AI commentary.

PART IV

Running the platform.

Five sections covering the operational substrate: state machines, tooling, runbooks, security, observability and cost. Every dataset shipped depends on these working.

23 · 24

Workflows &
tooling

25

Security &
compliance

26

Observability

27

Cost & capacity

Every record has a state machine. No exceptions.

Long-running business records (opportunities, briefs, builds, runs, contracts, deliveries, DSARs) move through explicit, audited state machines. State transitions emit events; events drive automation; automation never bypasses an explicit state. This is the difference between an operations platform and a CRM with side-effects.

Buyer opportunity state machine

```
new → qualifying → scoping → feasibility → pilot_quoted → pilot_active →
pilot_delivered → full_quoted → full_active → delivered → on_subscription |
closed_won | closed_lost
```

Supplier opportunity state machine

```
new → qualifying → nda_signed → sample_received → feasibility_done → pilot_active →
pilot_done → full_active → live → paused | terminated
```

Build state machine

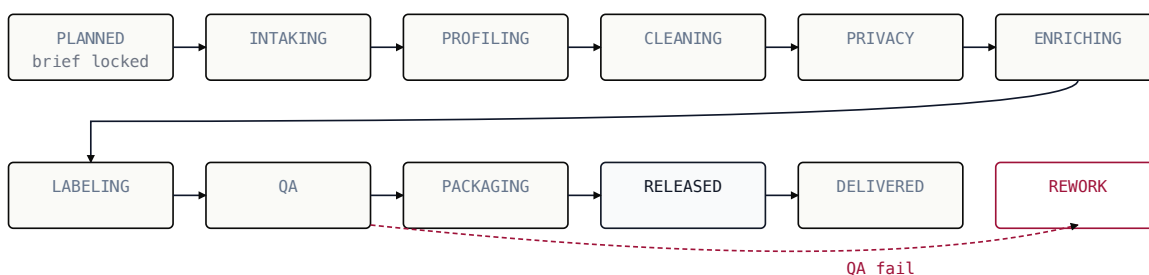


FIG 23.1 — BUILD STATE MACHINE; QA FAIL RE-ROUTES TO REWORK AND RE-ENTERS FROM THE APPROPRIATE STAGE.

FIG.
23.1

Other key state machines

- **Run** — `queued → running → succeeded | failed | cancelled` ; with retry counter, signal-aware.
- **Label batch** — `queued → in_review → in_adjudication → closed | requested` .

- **Contract** — drafting → awaiting_buyer → awaiting_supplier → signed → active → renewed | terminated .
- **Delivery** — scheduled → preparing → ready → sent → downloaded → accepted | disputed .
- **DSAR** — received → identity_verified → impact_assessed → propagating → completed ; SLA-tracked.

Automation patterns

Each state transition emits a typed event consumed by automation: notifications, downstream job triggers, billing events, audit log entries. Operators see an **activity** column on every record showing the state stream in human-readable form.

Tooling that operators reach for, not engineers.

Internal tooling exists to put platform power in the hands of operators without requiring them to write code, while still being scriptable for engineers when the situation demands it. The default is the console UI; the escape hatch is the CLI; the long-tail is direct database access under audit.

The Caudals CLI

A single binary `caudals` wraps the most common operations:

COMMAND	PURPOSE
<code>caudals build plan <brief></code>	Generate a build plan from a brief or template; outputs a YAML manifest for review.
<code>caudals build run <plan></code>	Submit a build plan to Dagster; tracks the run.
<code>caudals build replay <build_id></code>	Re-run a build from its manifest; verifies hash equality.
<code>caudals lineage trace <asset></code>	Pretty-print the lineage graph for an asset version.
<code>caudals license check <build></code>	Compute composed permitted-uses for a build before running it.
<code>caudals pii scan <asset></code>	Run the PII detector and dump a structured report.
<code>caudals dataset publish <version></code>	Promote silver to gold with the release manifest.
<code>caudals delivery sign <artefact></code>	Sign a packaged artefact for delivery.
<code>caudals dsar propagate <ticket></code>	Propagate a deletion/erasure request through derivative datasets.

COMMAND	PURPOSE
caudals fixture seed	Seed dev / staging with deterministic fixtures.

Runbooks (canonical)

- **R-01 · Onboard a new supplier asset.** NDA → intake → profile → feasibility memo.
- **R-02 · Open a buyer brief.** Capture → qualify → map to template → quote.
- **R-03 · Run a pilot build.** Plan → execute → QA → deliver → debrief.
- **R-04 · Promote pilot to full build.** Reuse plan with full-volume parameters; recompute license.
- **R-05 · Handle a QA failure at G-7.** Triage → root-cause → rework path → re-enter pipeline.
- **R-06 · Process a DSAR.** Verify → impact-assess → propagate → notify buyers.
- **R-07 · Rotate a signing key.** Generate → publish → rotate → deprecate.
- **R-08 · Restore from disaster.** RPO/RTO targets, restore order, verification checklist.
- **R-09 · Release a refresh.** Pull deltas → rebuild → QA → ship to subscribers.
- **R-10 · Deprecate a dataset version.** Notice → sunset window → tombstone.

Internal escalation surface

The console exposes a top-level *Escalations* view where any operator can flag: a suspected privacy incident, a contested provenance finding, a failing supplier delivery, a buyer dispute, or a security event. Escalations route to on-call operations and trigger the appropriate runbook automatically.

A posture regulators recognise.

Caudals' commercial credibility depends on its security and compliance posture from day one. The platform is built to a level compatible with SOC 2 Type II, ISO 27001, GDPR, CCPA/CPRA and EU AI Act Article 10 documentation, even before formal certifications are pursued. The section below describes the controls; certification timing is in §30.

Identity & access

- **Internal SSO** with mandatory MFA; phishing-resistant factors required for production roles.
- **Role-based access control** with least-privilege roles enforced at the database (RLS), service (server actions), and infrastructure layers (Tailscale ACLs).
- **Just-in-time elevation** for production database access; every elevation is recorded with reason and is time-bounded.
- **Buyer-side SSO + SCIM**; supplier-side SSO available on request.
- **Service-account hygiene**: short-lived credentials, rotated, never shared.

Data protection

- **Encryption in transit**. TLS 1.3 mandatory on every public and internal endpoint.
- **Encryption at rest**. Object store, Postgres, lakehouse tables; KMS-managed keys with documented rotation.
- **Field-level encryption** for sensitive columns where the threat model demands.
- **Customer-managed keys** available for enterprise buyers with regulatory requirements.
- **Backups**: encrypted, off-region, restore-tested; RPO 24h, RTO 4h for OLTP.
- **Bronze layer immutability**: object lock with retention period prevents tampering or premature deletion.

Network & infrastructure

- **Public surface area** is minimised: only the marketing site and approved API paths are reachable.
- **Admin access** via Tailscale only; public 22/tcp closed.
- **Egress controls**: outbound from operations VPC limited to allow-listed destinations; egress audited.
- **Secrets management**: centralised vault; no secrets in code, CI logs, or operator chat.

- **Vulnerability management:** Dependabot + automated container scans + scheduled penetration tests.
- **Incident response:** documented runbooks (R-08, security-specific R-11+), retainer with external IR firm.

Compliance & governance

FRAME	CAUDALS POSTURE
GDPR	Lawful basis tracked per asset; DPIA template; DSAR pipeline (R-06); DPO designated when required; data-processing register maintained.
CCPA / CPRA	Consumer-rights pipeline shared with GDPR; sale/sharing disclosure; deletion propagation through derivatives.
EU AI Act Art. 10	Per-dataset documentation: governance, bias testing, representativeness, sources; aligned with the dataset card and datasheet packaged at delivery.
SOC 2 Type II	Controls aligned to Trust Services Criteria from day one; pursue audit at year 2 when revenue justifies.
ISO 27001	ISMS scoped to platform; risk register maintained; pursue certification with SOC 2.
HIPAA	Optional posture for healthcare engagements: BAAs, segregated environment, encryption controls.
C2PA	Content credentials emitted on image / video / audio gold artefacts.
SPDX / CDLA	Standard license identifiers attached to every dataset version.

Audit & transparency

Every privileged action (data read, license change, key access, deletion, export) writes an `audit_event`. Audit retention is at least the longest contractual retention plus one year. Buyers and suppliers can request audit excerpts scoped to their own records and contracts.

If a build is failing, we see it before the buyer does.

The platform's observability posture must answer three classes of question fast: *is the platform healthy, is this build healthy, and did anything happen to this dataset version that matters*. Each question maps to a dedicated stack and surface.

Three classes of telemetry

CLASS	WHAT IT COVERS	STACK
Platform health	API latency, error rate, queue depth, GPU utilisation, storage capacity, container health.	OpenTelemetry → Prometheus + Grafana + Tempo + Loki + Sentry.
Pipeline health	Per-build progress, gate pass rate, run duration, retry rate, cost per build, reviewer queue depth.	Dagster events → OpenLineage → Marquez + Caudals dashboards.
Dataset events	Version creation, license change, refresh, DSAR propagation, delivery acceptance, deprecation.	Internal event bus → audit log + console activity feeds.

Service-level objectives (initial)

SERVICE	SLI	SLO
Public APIs	p95 latency	≤ 250 ms
Internal console	p95 page load	≤ 800 ms
Job queue	queue admit time	≤ 30 s p95
Pipeline	build replay equality	100% on weekly sample
Delivery	signed-URL availability	99.9% / month
Webhook receivers	processed within 5 min	99.5% / month

Alerting policy

Alerts are routed to PagerDuty / on-call rotation when they affect buyer-visible delivery, supplier-visible portal, or audit/compliance invariants. Internal-only platform issues route to a non-paging channel during business hours unless they threaten an SLO.

Build-cost telemetry

Every Dagster run reports compute time, GPU minutes, storage bytes read/written, and external API costs (LLM enrichment, third-party data, geocoding, etc.). Cost per build is a first-class metric on the build detail screen and rolls up to engagement-level margin reports.

Margin discipline at build level, not company level.

Dataset operations is a margin-sensitive business. A single sloppy build — one without cost ceilings, careless GPU usage, or unchecked LLM-enrichment spend — can erase the margin on the engagement. Cost discipline is therefore enforced at planning time, not after-the-fact.

Cost ledger

- Every Dagster op reports compute, storage, network and external-call costs.
- Costs accrete to a `build_id`, an `engagement_id`, and a `customer_id`.
- The console exposes *budget envelopes* per build with hard ceilings.
- Crossing 80% triggers a soft alert; crossing 100% requires explicit override.
- LLM and external API spending have separate sub-budgets; they cannot consume the whole envelope silently.

Capacity model

RESOURCE	BOTTLENECK	MITIGATION
CPU workers	Cleaning + profiling on TB tabular.	Polars / DuckDB scale-up; Spark only beyond TB.
GPU pool	CV / audio inference, embeddings, assistant labeling.	Burst-cloud capacity; on-prem baseline; pre-empt-able for non-critical.
Object storage	Multi-version retention + bronze immutability.	Tiered storage; cold tier for bronze older than N months.
Postgres	OLTP + lineage queries.	Read replicas for analytics; partitioning for audit_event.
Reviewer hours	Labeling SLA.	Workforce scheduling; pre-label automation; active learning.
External LLM API	Per-build enrichment cost.	Local OSS models for high-volume; per-prompt budget enforcement.

Pricing-aware planning

The build planner consumes the pricing table for each operator (compute per minute, storage per GB-month, GPU per hour, LLM per thousand tokens) and produces a cost estimate alongside the plan. This estimate is what operators quote against in §02 deliverables (feasibility memo, pilot quote, full-build quote). When actuals deviate from estimates by $> 15\%$, a margin retrospective runs with the operator and the head of operations.

PART V

Implementation plan.

Seven sections that turn the blueprint into a build sequence: the data model, the API surface, the roadmap, the team, the risks, a glossary, and the references that informed every decision in this document.

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Schema & APIs

30

Roadmap

31 · 32

Team & risks

33 · 34

Glossary & refs

The schema is the operations spine.

The list below is the canonical core of the platform's relational schema. Naming is final; ID prefixes match the records; foreign-key edges are indicative. Migrations land in `supabase/migrations/*` and never bypass review.

Domain map

Organizations & people

- `organization` — supplier, buyer, partner, internal.
- `contact` — named individuals at orgs, with roles and signing authority.
- `operator` — Caudals staff with role and skill profile.
- `reviewer` — labelers / adjudicators (in-house, partner, SME).

Demand & supply

- `buyer_opportunity` — pipeline record from inbound through close.
- `supplier_opportunity` — pipeline record for supplier acquisition.
- `dataset_brief` — structured buyer requirement.
- `supplier_asset` — declared data asset with rights metadata.

Rights & consent

- `contract` — signed legal agreement (supplier, buyer, NDA).
- `license_clause` — structured permitted-use granted by a contract.

Datasets & lineage

- `dataset` — logical dataset family (a "SKU").
- `dataset_version` — concrete materialised version with hash, size, license, scorecard.
- `dataset_partition` — physical partitions in the lakehouse.
- `lineage_event` — OpenLineage event reference.
- `manifest_artifact` — build manifests, dataset cards, datasheets.
- `pii_map` — per-version PII detection & treatment record.

Catalogue & commerce

- `catalogue_listing` — public/partner listing entry.
- `private_offer` — targeted offer to a specific buyer.
- `quote` — priced proposal for a build / subscription.
- `subscription` — recurring delivery program.
- `delivery` — specific delivery instance with channel + receipt.
- `invoice`, `payment`, `payout` — commerce records.
- `revenue_share` — supplier-side accruals.

- `consent_record` — subject-level consent attached to records or assets.
- `dsar_request` — subject-rights ticket with propagation tracking.

Pipeline state

- `build` — orchestration record for a buyer/supplier engagement.
- `build_plan` — serialized YAML manifest of the plan.
- `run` — a Dagster run instance against a `build_plan`.
- `gate_event` — pass / fail decisions at each gate.
- `label_batch` — unit of work in the labeling fabric.
- `qa_report` — quality scorecard with dimensions and verdict.

Operations & audit

- `audit_event` — cross-domain auditable action log.
- `cost_entry` — granular cost ledger line.
- `alert` — operations alert with state.
- `integration` — per-org integration credentials (encrypted).
- `signing_key` — platform / per-customer keys for delivery signing.

Cross-cutting fields

Every record carries: `id` (ULID with a 2-char prefix per type), `created_at`, `updated_at`, `created_by`, `org_id` (RLS scope), `state` (per state machine), `deleted_at` (soft delete with retention policy). Tenancy is enforced at row level via Postgres RLS.

Selected DDL — the load-bearing tables

```
-- supplier_asset: a declared data asset under a supplier contract
create table supplier_asset (
  id                text primary key,          -- "sa_" + ULID
  org_id            text not null references organization(id),
  name              text not null,
  modality          text not null check (modality in
    ('tabular', 'text', 'image', 'video', 'audio', 'geospatial', 'timeseries', 'document')
  ),
  declared_volume   jsonb,                    -- {records|bytes|hours...}
  refresh_policy    text not null,             -- one_shot|scheduled|on_event|perpetual
  sensitivity       text not null,             -- public|confidential|pii|special
  rights_summary    jsonb not null,            -- structured permitted-uses
  state             text not null default 'declared',
  created_at        timestampz not null default now(),
  updated_at        timestampz not null default now(),
  created_by        text references operator(id)
);

-- license_clause: structured permitted use granted by a contract
create table license_clause (
  id                text primary key,          -- "lc_" + ULID
  contract_id       text not null references contract(id),
  asset_scope       jsonb not null,            -- which supplier_asset(s) it covers
  permits_train     boolean not null,
  permits_finetune  boolean not null,
  permits_eval      boolean not null,
  permits_inference_commercial boolean not null,
  permits_redistribute boolean not null,
  exclusivity       text not null default 'none', -- none|exclusive|category
  geo               text[] not null default '{{WW}}',
  term_starts_at    timestampz,
  term_ends_at      timestampz,
  share_alike       boolean not null default false,
  notes             text
);

-- dataset_version: concrete materialised version of a dataset family
create table dataset_version (
  id                text primary key,          -- "dv_" + ULID
  dataset_id        text not null references dataset(id),
  version_label     text not null,             -- "v3", "2026.05"
  build_id          text references build(id),
  manifest_uri      text not null,             -- s3://...manifest.json
  content_hash      text not null,             -- sha256 of canonical artefact
  size_bytes        bigint,
  record_count      bigint,
  composed_permits  jsonb not null,            -- intersection of input licenses
  qa_score          numeric(4,3),
  state             text not null default 'draft',
  released_at       timestampz,
  released_by       text references operator(id),
```

```
signed_by      text      -- signing_key.id
);
```

Three audiences, three API styles.

The platform exposes three distinct API styles, each for a defined consumer. Mixing styles between consumers is forbidden; consumers see one API, contracts are tested at the boundary, and contract drift is caught in CI before deploy.

API styles

STYLE	AUDIENCE	REASON
Server actions	Internal console (operator UI).	End-to-end type safety with the React client; lowest friction.
tRPC	Buyer workspace, supplier portal.	Type-safe with versioned schemas; aligns with internal record model.
REST + signed	Partner integrations, programmatic delivery, webhooks.	Standard, language-neutral, easier for buyer engineering teams to consume.

Selected REST endpoints (buyer-facing)

```
# Datasets
GET    /v1/datasets           # list catalogue / authorised
GET    /v1/datasets/{id}      # dataset family detail
GET    /v1/datasets/{id}/versions  # versions for this dataset
GET    /v1/datasets/{id}/versions/{ver}  # specific version detail
GET    /v1/datasets/{id}/versions/{ver}/sample  # preview (NDA-gated for private)
POST   /v1/datasets/{id}/versions/{ver}/access  # request access to a private listing

# Briefs
POST   /v1/briefs         # submit a custom-build brief
GET    /v1/briefs/{id}    # brief detail / progress
POST   /v1/briefs/{id}/approve_quote  # buyer-side approval

# Deliveries
GET    /v1/deliveries     # list buyer's deliveries
GET    /v1/deliveries/{id}  # delivery detail + receipt
POST   /v1/deliveries/{id}/accept  # countersignature
POST   /v1/deliveries/{id}/dispute  # raise an acceptance issue
```

```
# Subscriptions
GET    /v1/subscriptions                # active recurring deliveries
GET    /v1/subscriptions/{id}/refreshes # refresh history
POST   /v1/subscriptions/{id}/pause    # pause refresh program

# Webhooks (caudals -> buyer)
POST   {buyer-receiver}           # delivery.ready, refresh.published, dataset.de
```

Authentication & signing

- Bearer tokens for buyer/supplier APIs, scoped per organization.
- HMAC signatures on partner-emitted webhooks; replay window 5 min.
- Caudals webhooks signed with rotating Ed25519 keys; receivers verify against published key set.
- Signed-URL deliveries use short-lived tokens with audit-logged retrieval.

Versioning & deprecation

API surfaces are versioned in path (`/v1/...`). Breaking changes ship under a new version with a 6-month overlap. The deprecation calendar is published; deprecated endpoints emit a `Sunset` header and a deprecation event in webhooks. No silent breaking changes ever.

Internal-only endpoints

Operations endpoints (job submission, lineage queries, license-policy evaluation, DSAR propagation) are reachable only from within the operations VPC and require a Caudals service identity. They are not documented in any external surface.

Operator-grade in 90 days. Self-serve by month 12.

The roadmap is sequenced to put the operations spine on its feet first, populate the catalogue with real, sold datasets next, and ship self-service surfaces only after operations are reliable. Dates are targets relative to Day 0 = adoption of this blueprint.

M0 · Day 0–30 — Foundations

- Adopt the schema (§28); migrate any existing admin-side records.
- Stand up bronze / silver / gold buckets with lakeFS atop existing object store.
- Stand up Dagster orchestrator with three reference assets (intake, profile, qa).
- Implement license-clause vault and the composition algebra (§17).
- Ship the operator console v0: leads, suppliers, buyers, builds (read-only deep links).
- Wire OpenLineage emission from existing pipelines into Marquez.

M1 · Day 30–90 — Operator-grade

- Pipeline gates G-1...G-7 enforced; rebuild & replay CLI working.
- PII detector pool with Presidio + spaCy + custom recognisers.
- Cleaning operator library v1 (~30 operators) with deterministic-output tests.
- Label Studio embedded for tabular / text; CVAT for images; reviewer queue with adjudication.
- QA scoring engine v1 with composite Q-score and exception report.
- Packaging targets: Parquet, JSONL, COCO, HF Datasets, signed-URL delivery.
- Operator console v1: full CRUD, command palette, audit overlays, build detail screen.
- **North-star milestone:** 5 concurrent builds, end-to-end with full lineage and licensed delivery.

M2 · Day 90–180 — Catalogue & modalities

- Catalogue listing model + private-offer model + sample-preview gating.
- Modality coverage extended: video, audio, geospatial.
- Active-learning loop wired with FiftyOne Brain / Lightly-style sampling.
- Cleanlab pass integrated into the QA stage.
- Subscription delivery model with delta manifests.

- Buyer workspace v0: read-only delivery + scorecard view.
- Supplier portal v0: asset declaration, sample upload, build participation.

M3 · Day 180–365 — Self-service & commercial

- Public catalogue at `/catalogue` (curated subset).
- Buyer brief intake from public web; auto-routed to operator queue.
- Buyer workspace v1: subscriptions, integrations, billing.
- Supplier portal v1: revenue share, payouts via Stripe Connect.
- Document & time-series modality coverage at production quality.
- SOC 2 Type I audit underway; ISO 27001 ISMS scoped.
- EU AI Act Article 10 documentation auto-generated per dataset.
- Croissant manifest emission default; HF mirror channel for public assets.

M4 · Beyond 12 months

- SOC 2 Type II completed; HIPAA-eligible environment for healthcare.
- Self-serve catalogue purchase flow for low-friction asset classes.
- Programmatic-labeling and weak-supervision flows productised.
- Cross-region delivery (US east + EU west); customer-managed keys.
- Marketplace network effects: supplier ↔ buyer matching by signal.

A small team. Clear ownership.

The first 12 months are deliberately small-team. Roles below are function-level; one person can fill multiple. The RACI matrix is what matters: every load-bearing area of the platform has exactly one accountable owner.

Function map (12-month team)

FUNCTION	RESPONSIBILITY	HEADCOUNT TARGET
Founder / CEO	Strategy, sales, fundraising, supplier outreach.	1
Founder / CTO	Architecture, security, platform sequencing.	1
Platform engineer	App + infra: console, APIs, deployment.	1–2
Data engineer	Pipeline core: orchestrator, lakehouse, lineage.	1–2
ML / data scientist	QA scoring, active learning, profiling, embeddings.	1
Operations lead	Build delivery, supplier coordination, buyer success.	1
Labeling lead	Reviewer fabric, ontology governance, partner workforce.	1 (year 2)
Privacy / compliance lead	Rights, consent, DSARs, certification programs.	1 fractional then full
Sales / partnerships	Buyer demand, supplier acquisition.	1 (year 2)
Designer	Console + buyer/supplier UX, brand.	1 fractional

RACI on load-bearing areas

AREA	R	A	C	I
Architecture	CTO	CTO	Platform, Data eng	Founders

AREA	R	A	C	I
Pipeline core	Data eng	CTO	ML, Ops	Founders
Console UX	Platform eng + Designer	CTO	Ops	Founders
Privacy posture	Privacy lead	CEO	CTO, Ops	Buyers
Build delivery	Ops lead	CEO	Data eng, Privacy	Buyers
Quality scoring	ML	CTO	Ops	Buyers
Catalogue listings	Ops + Designer	CEO	Sales, Privacy	Suppliers
Commercial ops	CEO	CEO	Ops, Privacy	Founders
Security	CTO	CTO	Privacy	Buyers
Compliance certifications	Privacy lead	CEO	CTO	Buyers

The honest register.

Every line in this register is a known way the platform can fail commercially, legally, technically or operationally. Each carries an owner. The register is reviewed monthly; mitigations move from "planned" to "in place" with audit evidence.

ID	RISK	MITIGATION	SEVERITY	OWNER
R-01	Privacy incident	P-05 ingest-time PII; immutable bronze; KMS-controlled token vault; staff training; IR retainer.	CRITICAL	Privacy
R-02	Licensing dispute	License clauses as code; intersection at plan time; signed contracts attached; audit retention.	CRITICAL	CEO/Privacy
R-03	Buyer rejects on QA	Sample previews + scorecards before quote; pilot before full build; 7-day acceptance window with clear dispute path.	HIGH	Ops
R-04	Supplier withdraws data	Termination clauses; tombstone propagation; subscription deltas reversible; alternative-supplier mapping for core SKUs.	HIGH	CEO
R-05	Build cost overrun	Cost ledger + envelopes; LLM sub-budgets; margin retros > 15%.	HIGH	Ops
R-06	Compute capacity shortfall	Burst-cloud GPU; non-critical pre-emption; queue prioritisation by build SLA.	MEDIUM	CTO
R-07	Reviewer quality drift	Gold questions; agreement metrics; Cleanlab pass; reviewer cohort rotation.	MEDIUM	Labeling
R-08	Lineage gaps	OpenLineage default-on; CI replay tests; nightly lineage-completeness audit.	MEDIUM	Data eng

ID	RISK	MITIGATION	SEVERITY	OWNER
R-09	Schema drift between modules	Generated types from migrations; contract tests in CI; PR-level migration review.	MEDIUM	Platform
R-10	Vendor lock-in (lakeFS, Dagster, etc.)	Choose vendor-neutral protocols (OpenLineage, Iceberg, Croissant); evaluate alternatives yearly.	MEDIUM	CTO
R-11	Regulatory change (EU AI Act, etc.)	Per-asset documentation auto-generated; legal counsel on retainer; quarterly posture review.	HIGH	Privacy
R-12	Reputational risk from sourced data	Source allow-list; supplier KYC; refusal categories documented; public ethics policy.	HIGH	CEO
R-13	Single founder/key-person dependency	Cross-train; runbooks (R-01..R-10); BCP plan; bus-factor in security reviews.	MEDIUM	CEO
R-14	Buyer security review delays	Public security page (M3); SOC 2 in flight; standard DPA; security questionnaire library.	MEDIUM	Privacy
R-15	Synthetic-data utility / privacy gap	Conservative defaults; utility-vs-privacy report at every synthetic build; no silent blending with real.	MEDIUM	ML

Vocabulary used throughout the platform.

Naming is part of the architecture. Every Caudals operator, engineer and document should use these terms identically. New vocabulary is proposed via PR against this glossary.

Asset

A first-class artefact: a `supplier_asset` (input), a `dataset_version` (output), or a `manifest_artifact`. Every asset has a hash, a license posture, and a lineage manifest.

Bronze / Silver / Gold

Medallion layers: raw + sealed (bronze), cleaned/conformed/PII-handled (silver), ML-ready packaged + scored (gold).

Build

A scoped engagement that turns one or more `supplier_assets` and a `dataset_brief` into one or more `dataset_versions`, governed by a `build_plan`.

Build plan

A YAML manifest naming the operators, parameters, ontology, gates, and license posture for a build. Versioned in git.

Catalogue listing

A public or partner-only entry exposing a `dataset_version` family for purchase, with sample preview, scorecard, license tier, and price.

Composed permits

The intersection of permitted-uses across input licenses, computed at build plan time; the

Manifest

A signed JSON describing a packaged dataset: version, content hash, schema, license, lineage references, signing key.

Operator (Caudals)

A staff role with platform access: data engineer, operations lead, privacy lead, etc. Distinct from "operator" in the pipeline sense.

Operator (pipeline)

A reusable, typed transformation in the cleaning / enrichment / labeling library; a Dagster op with a contract.

PII map

A per-version structured record of detected PII locations and the treatment applied to each.

Programmatic labeling

Generation of labels through heuristics, rules, classifiers and LLMs (labeling functions), unified by a label model.

QA scorecard

The composite quality artefact attached to every released `dataset_version`, exposing component scores and a release verdict.

Reproducible build

maximum set of uses the output can support.

DSAR

Data Subject Access Request — including erasure, portability, and rectification requests. Tracked end-to-end through derivative datasets.

Gate

A contract between two pipeline stages. Promotion to the next stage requires a passing gate evaluation and a recorded sign-off.

Gold question

A pre-labeled item secretly inserted into a reviewer's queue to measure ongoing reviewer accuracy.

HITL

Human-in-the-loop — labeling strategies that put human reviewers in the loop with model assistance.

Lineage event

An OpenLineage-compliant record of a transformation: inputs, outputs, code SHA, parameters, status, facets.

License clause

A structured permitted-use record granted by a contract: train, fine-tune, eval, infer commercial, redistribute, geo, exclusivity, term.

A build whose dataset_version is byte-identical to its prior release when re-run from the manifest at the same git SHA. Verified weekly in CI.

RLS

Row-level security — Postgres policy enforcement scoped per organization; the primary tenancy boundary.

SKU (dataset)

A logical dataset family with a stable identity, multiple versions, and a license tier; the unit of catalogue listing.

Subscription delivery

A recurring delivery program where successive dataset_versions are emitted to the buyer with delta manifests.

Tombstone

A marked-deleted dataset_version with retained metadata and a deletion-required notice for downstream consumers.

Weak supervision

Snorkel-style labeling: heuristics and small models vote; a label model resolves conflicts to produce probabilistic labels.

What this blueprint stands on.

Public references and standards that informed this document, grouped by domain. Internal Caudals documents are listed first; the rest are public primary sources.

Internal Caudals references

- `AGENTS.md` — agent operating instructions (root of repository).
- `docs/product-specs/overview.md` — canonical product / business briefing.
- `docs/ARCHITECTURE.md` — live technical contract and deployment posture.
- `docs/DESIGN.md` — design system and visual governance.
- `docs/FRONTEND.md` — frontend implementation contract.
- `docs/TOOLS.md` — operational tooling and runbook.

Standards and specifications

- **OpenLineage Specification** (Linux Foundation AI & Data) — lineage event format adopted platform-wide.
- **MLCommons Croissant** — machine-readable dataset metadata interchange.
- **C2PA Technical Specification 2.x** — content credentials for media provenance.
- **Apache Iceberg & Lance** table formats — canonical lakehouse formats.
- **SPDX License List & CDLA** (Linux Foundation) — license identifier registry.
- **STAC** (SpatioTemporal Asset Catalog) — geospatial cataloguing standard.
- **Hugging Face Dataset Cards** — standard documentation pattern adopted at delivery.
- **W3C PROV** — provenance data model underpinning lineage thinking.

Regulatory frames

- **EU AI Act, Regulation (EU) 2024/1689**, especially Article 10 (training-data governance for high-risk AI).
- **GDPR (EU 2016/679)** and DPIA / DSAR provisions.
- **CCPA / CPRA** — consumer privacy posture for US engagements.
- **SOC 2 Trust Services Criteria 2017** with 2022 revisions.
- **ISO/IEC 27001:2022** ISMS requirements.
- **HIPAA Security & Privacy Rules** for healthcare engagements.

Foundational papers & bodies of work

- Gebru et al., *Datasheets for Datasets*, CACM, 2021.
- Mitchell et al., *Model Cards for Model Reporting*, FAT*, 2019.
- Northcutt et al., *Confident Learning*, JAIR, 2021 (Cleanlab).
- Ratner et al., *Snorkel: Rapid Training Data Creation with Weak Supervision*, VLDB, 2017.
- Carlini et al., series of papers on training-data extraction and benchmark contamination, 2021–2024.
- Stanford CRFM, *HELM* & *HEIM* evaluation reports.
- OpenDP and Tumult Analytics technical documentation on differential privacy in practice.

Reference platforms surveyed (non-endorsement)

Scale AI, Surge AI, Labelbox, V7 Darwin, Encord, Roboflow, Snorkel Flow, Cleanlab Studio, Galileo, Lightly, Hugging Face Hub, Voxel51 FiftyOne, Activeloop Deep Lake, Argilla, Label Studio, CVAT, doccano, Prodigy, Great Expectations, Soda Core, Pandera, ydata-profiling, whylogs, Microsoft Presidio, OpenDP, SDV, Gretel, Mostly AI, Qdrant, LanceDB, Weaviate, pgvector, lakeFS, DVC, Pachyderm, Marquez, Dagster, Temporal, Prefect, Airflow, Feast, MLflow, Unstructured.io, LlamaParse, AWS Textract, TorchGeo, GeoPandas, TimescaleDB.

One platform. One operations spine. One source of truth.

This document is the canonical technical specification for the Caudals platform. Updates land via PR against `docs/blueprints/` ; supersedes are recorded explicitly in §00.

VERSIONING

Document revisions follow semantic versioning. Breaking changes to architecture or schema require a major bump and a migration plan.

MAINTENANCE

Owned by Caudals Platform Engineering. Reviewed quarterly; triggered updates after major product or compliance changes.

DISTRIBUTION

Internal · do not distribute externally without explicit founder approval. Excerpts may be quoted in buyer security reviews.

CAUDALS

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